

CHY1008
Chemistry and Environmental Studies

Module 4

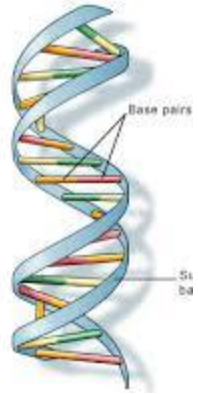
NANOMATERIALS

Engineering Materials

Dr. Sabeel M Basheer

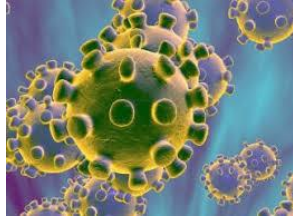
What is **Nano**??

Nano = 1 billion



2 nm

DNA



30 nm

Virus



2 μ m

Microns

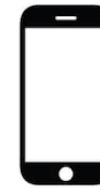
Bacteria



100 μ m
microns



2 mm

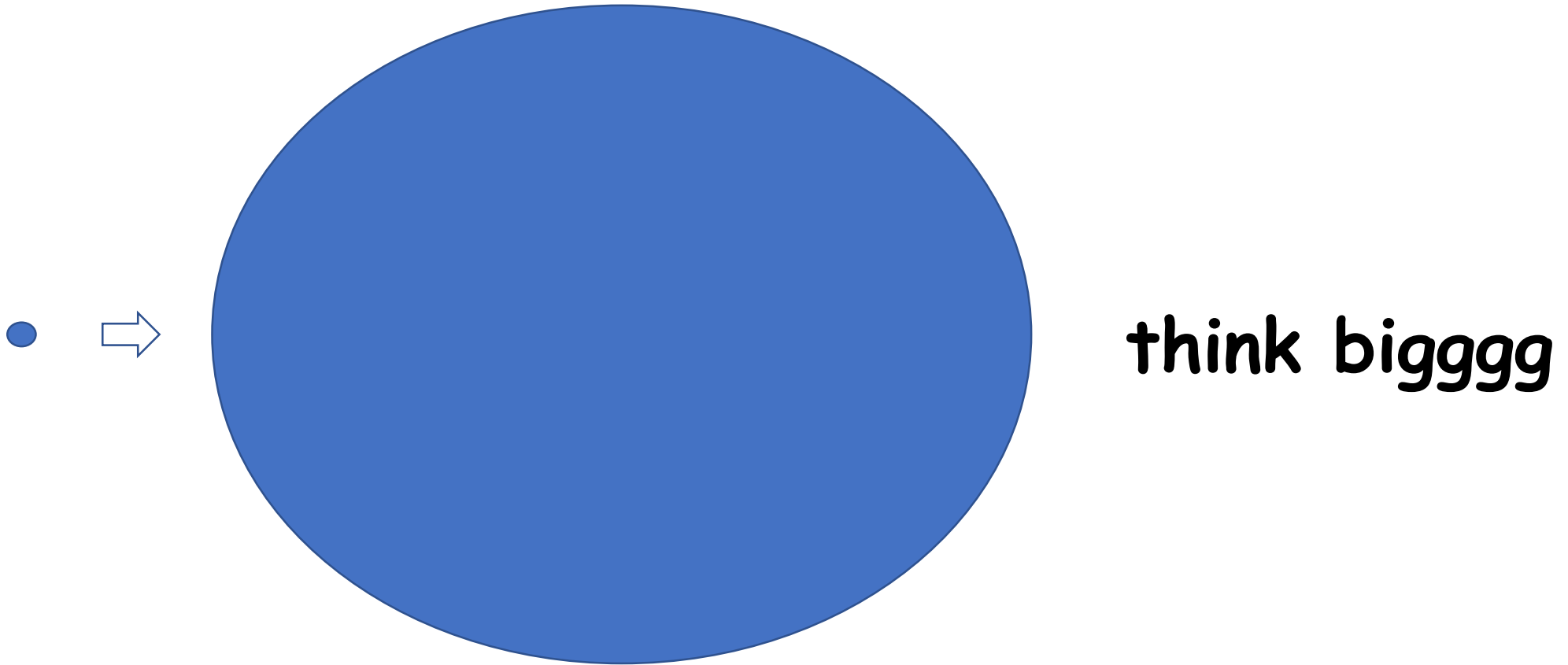


10 cm



1.5 – 2 m

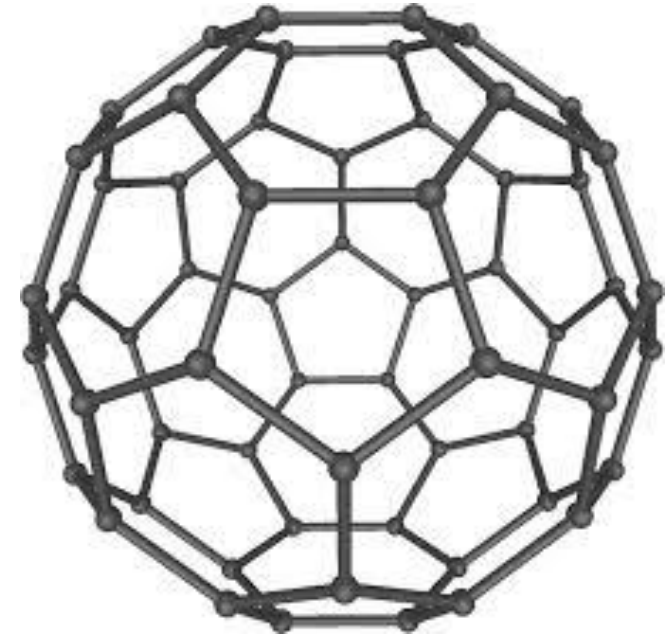
What is Nanotechnology ??



Nanotechnology is the collection of procedures for *manipulating matter on this scale* in order to build nanosized entities for useful purposes

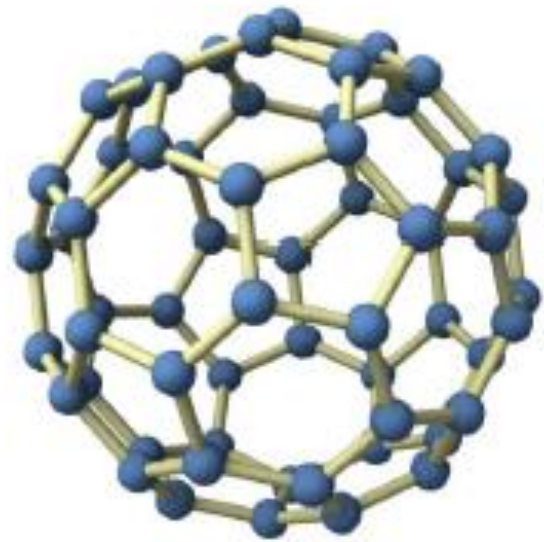
- ***The original version of nanotechnology occurred in *nature****
Ex: DNA, Photosynthesis, *Metallic NP, Metal Oxide NP*

The discovery of C_{60} and other related molecules, now collectively known as the **fullerenes**, helped usher in a new form of carbon and also a new branch of science—**nanotechnology**

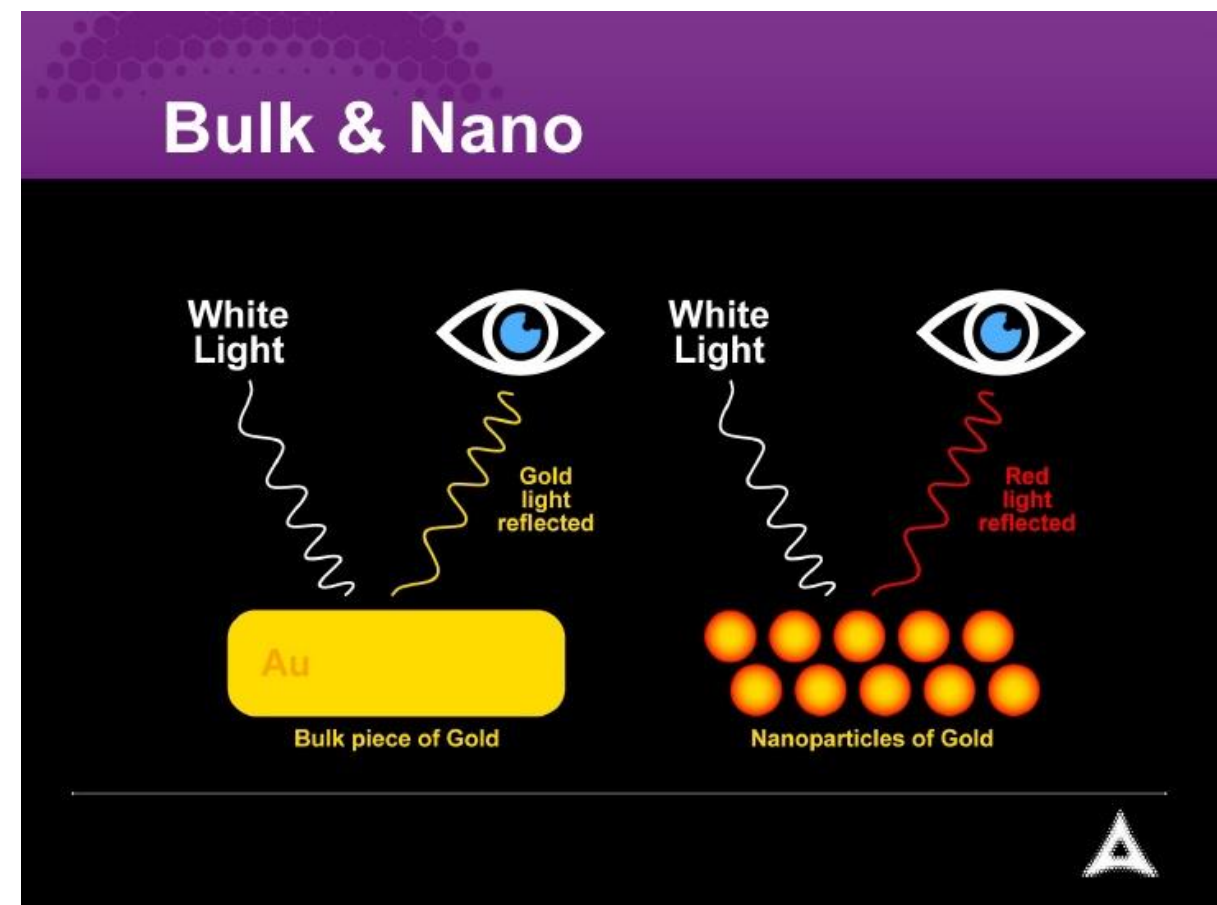
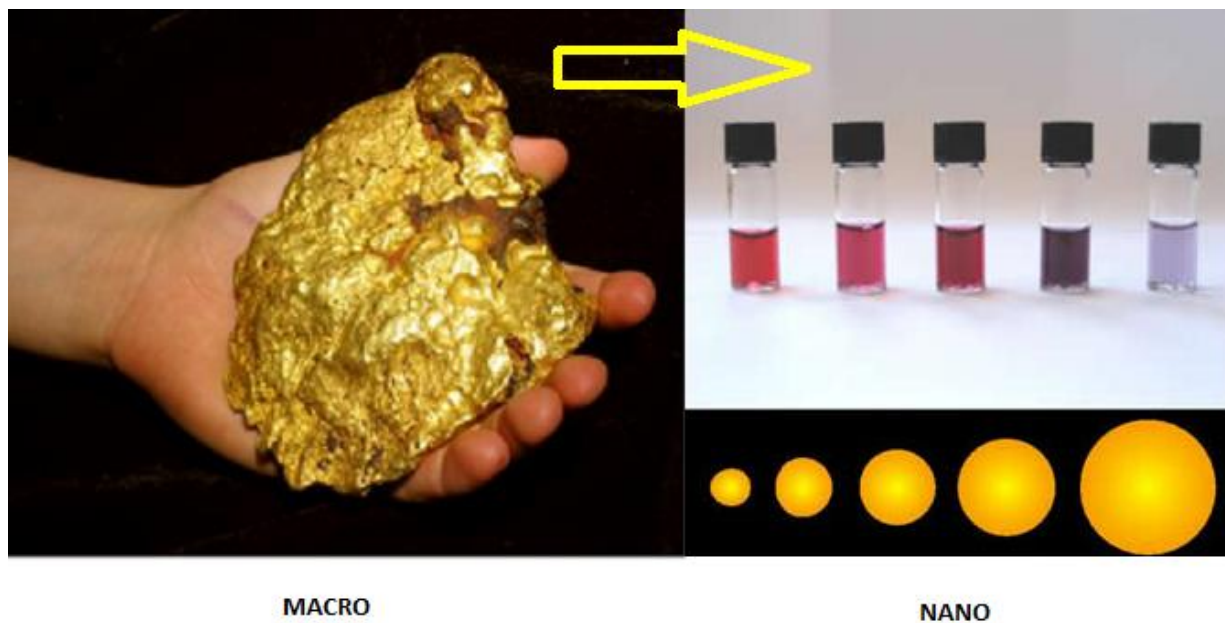


buckminsterfullerene

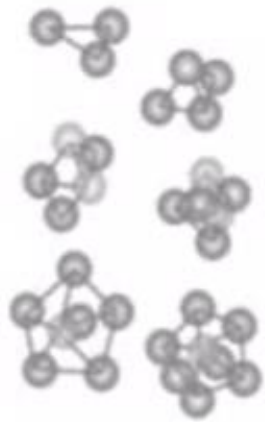
ratio
earth / football = ratio
football / fullerene



Bulk vs. Nano - materials



Classification of NP



Clusters
0D

Metal NP,
Metal Oxide NP,
Quantum Dots



Nanotubes, fibers and rods
1D

Nano wires,
Quantum wires,
Nano rods
Nano tubes



Films and coats
2D

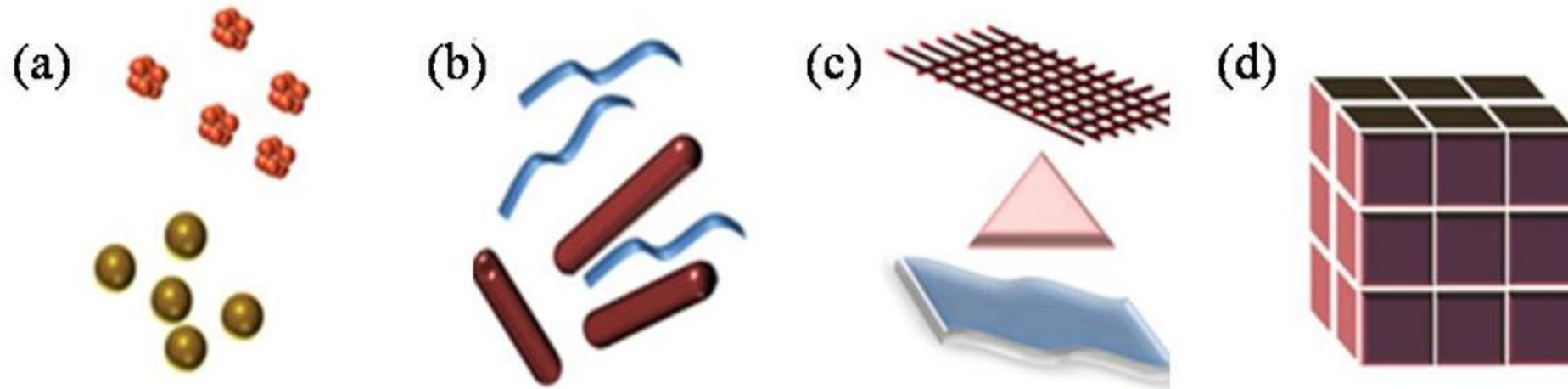
Thin Films



Polycrystals
3D

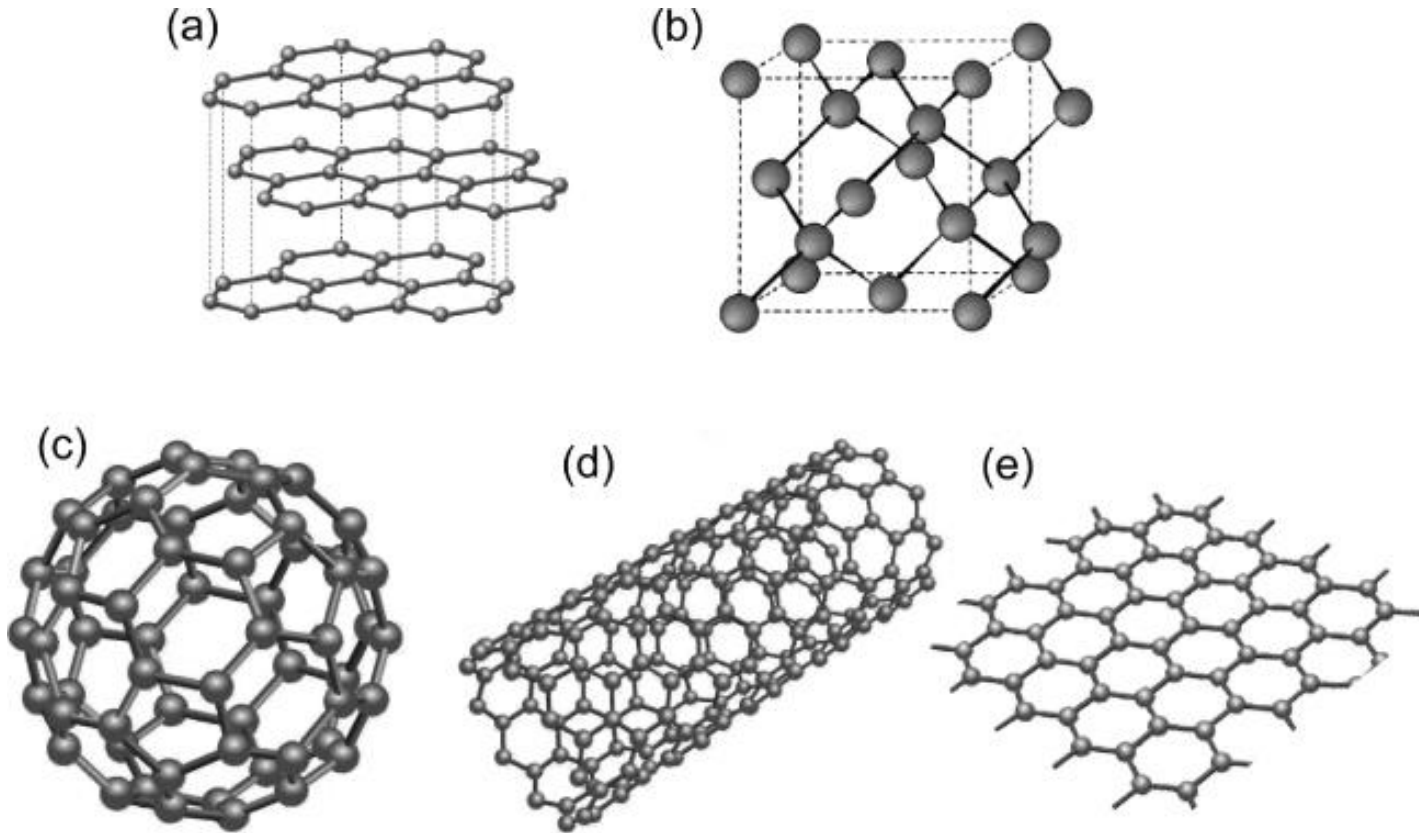
Bulk Nano Crystalline film
Nanocomposites

Examples of 0, 1, 2, 3-Dimensional Nanomaterials



Classification of Nanomaterials (a) 0D spheres and clusters, (b) 1D nanofibers, wires, and rods, (c) 2D films, plates, and networks, (d) 3D nanomaterials.

Examples of 0 to 3D Materials



Forms of carbon:

a) Graphite 3D

b) Diamond 3D

c) Buckminsterfullerene
“Buckyballs” 0D

d) Carbon nanotube 1D

e) Graphene 2D

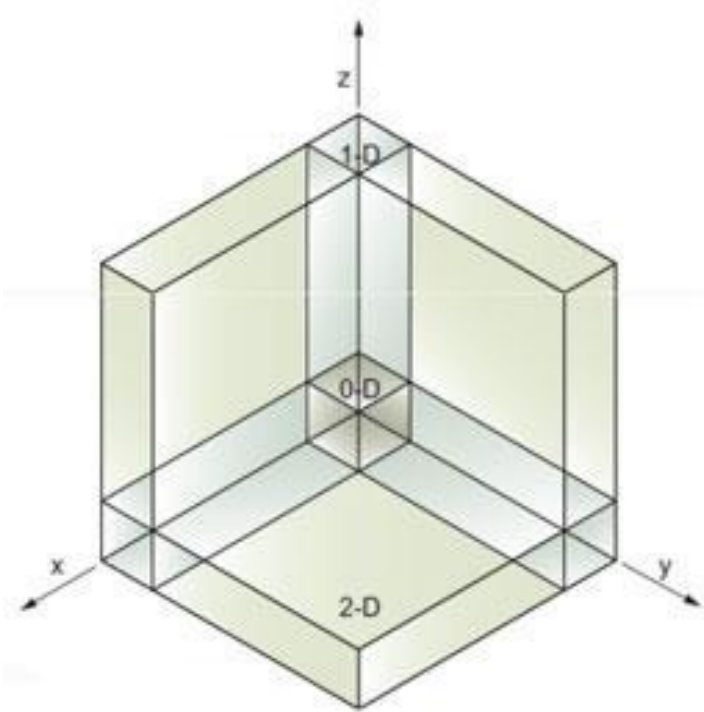
Bulk and nanoscale materials

The **key difference** between nanomaterials and bulk materials is that the **nanomaterials have their size in 1-100 nm** range at least in one dimension whereas the **bulk materials have their size above 100 nm** in all dimensions.

Classification (Macro scale)	Example	Nanoscopic Dimensions
Bulk (3D)	Anything you can see by eye	0
2D (nanosheet)	Graphene	1
1D (nanotube or nanowire)	Carbon nanotube	2
0D (nanoparticle)	Quantum dots (QDs)	3

Quantum Confinement Effect

Quantum confinement refers to the confinement of the electrons in nanoscale objects such as nanospheres, tubes, sheets, crystals.

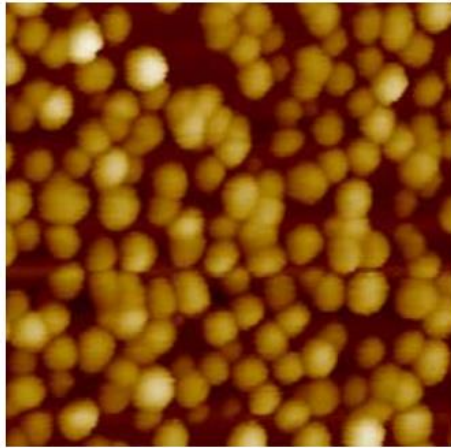


Dimension at macroscale	Electron confinement at nanoscale	Example
0D	3D	CdSe-QD
1D	2D	Carbon nanotube
2D	1D	Graphene
3D	0D	Micron sized particle

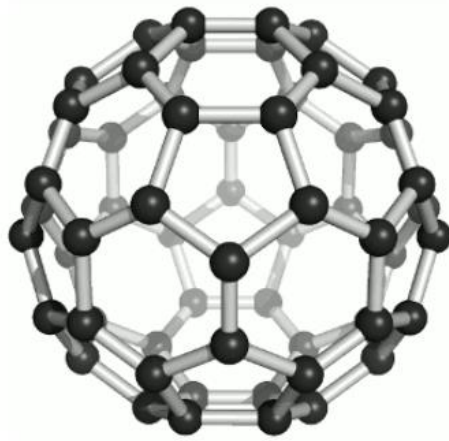
Example

Nanosphere is a zero-dimensional material.
Therefore electrons are confined in 3-Dimension.

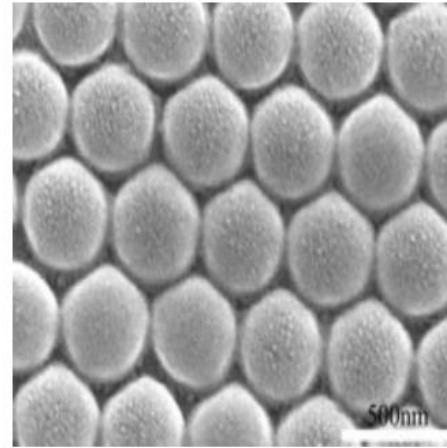
Examples of Nanomaterials



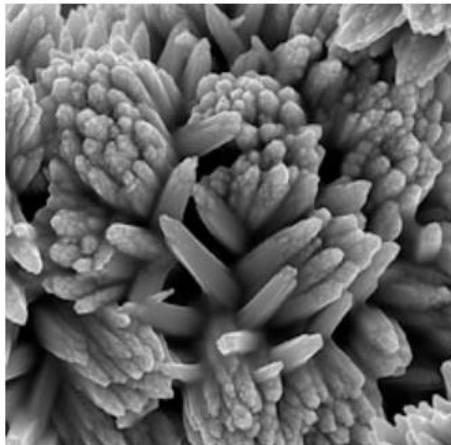
Au nanoparticle



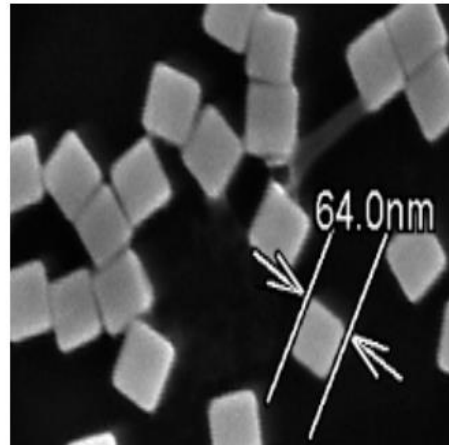
Buckminsterfullerene



FePt nanosphere



Titanium nanoflower



Silver nanocubes



SnO₂ nanoflower

Applications of Nanoparticles

- **Silver nanoparticles** have good **antibacterial properties**, and are used in **surgical instruments**, refrigerators, air-conditioners, water purifiers etc.
- **Gold nanoparticles** are used in catalytic synthesis of silicon nano wires, **sensors carrying the drugs** and in the **detection of tumors**.
- **ZnO nanoparticles** are used in **electronics**, ultraviolet (UV) light emitters, piezoelectric devices and **chemical sensors**.
- **TiO₂ nanoparticles** are used as photocatalyst and **sunscreen cosmetics** (UV blocking pigment).
- Antimony-Tin-Oxide (ATO), Indium-Tin-Oxide (ITO) nanoparticles are used in **car windows**, **liquid crystal displays** and in **solar cell** preparations.

Why **NANO** ?

At the nanoscale, strange things happen to materials
- their **properties** can change

Size Dependent Properties of Nanomaterials

Reactivity

Optical properties

Magnetic property

Electronic property

Thermal property

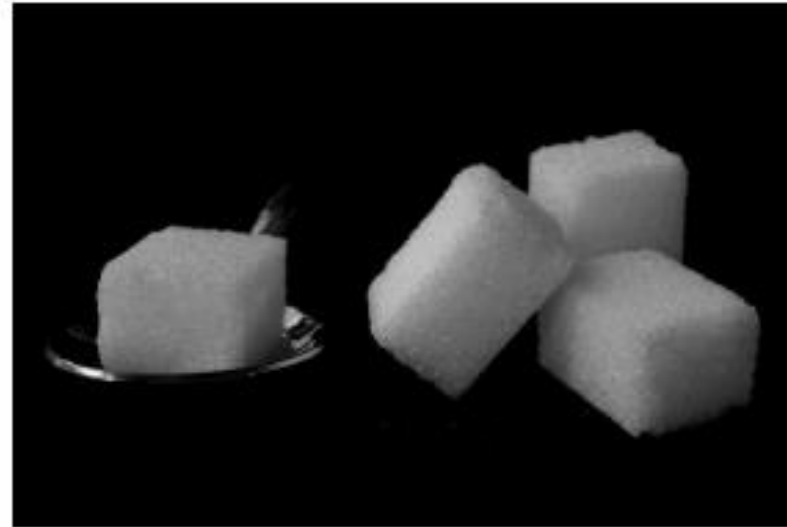
Size Dependent Properties of Nanomaterials

The various properties, which get tremendously altered due to the size reduction in at least one dimension are:

- a) Chemical properties: Reactivity; Catalysis.
- b) Thermal property: Melting point temperature.
- c) Electronic properties: Electrical conduction.
- d) Optical properties: Absorption and scattering of light.
- e) Magnetic properties: Magnetization.

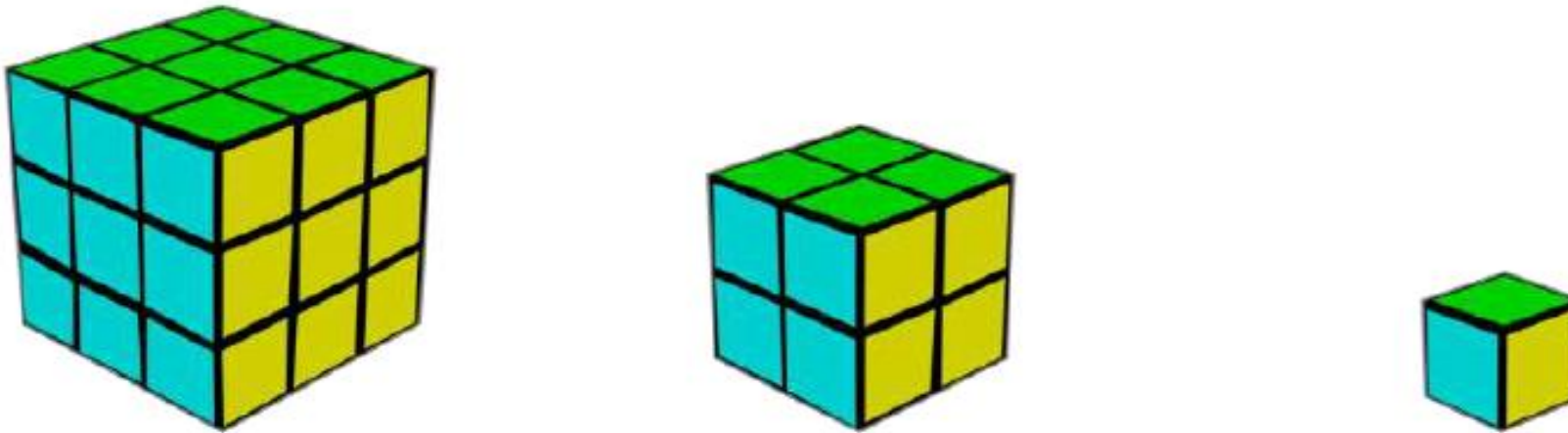
Chemical property: **Reactivity**

Which dissolves faster in water?



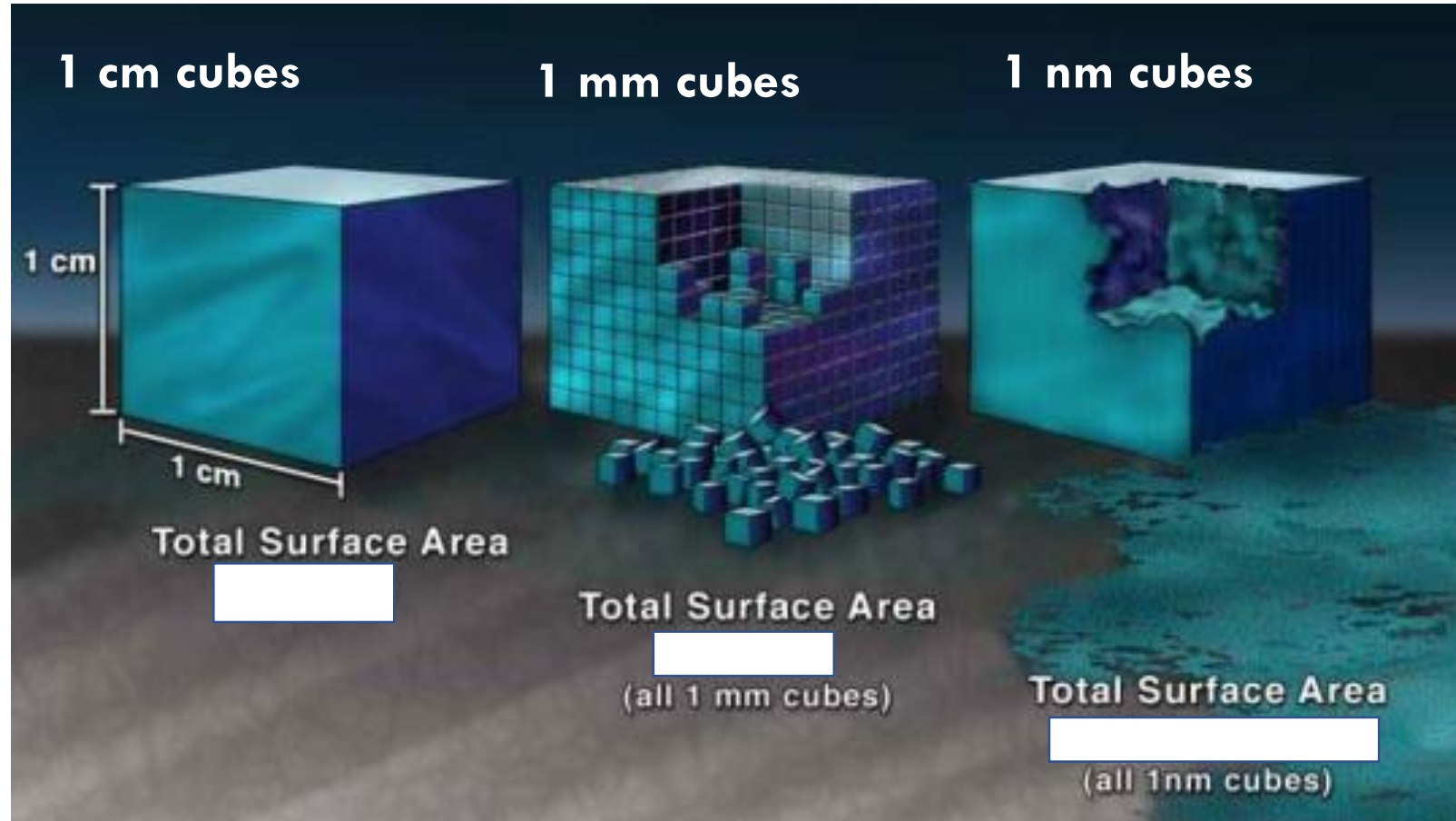
Chemical property: **Reactivity**

Surface Area to **Volume Ratio**



As surface area per mass of material increase greater amount of the material can come in contact with surrounding material thus affect the **reactivity**

Chemical property: **Reactivity**



Crushing a 1 cm particle into **nano** particles increases the surface area **thousands of times!!!**

Chemical property: **Reactivity**

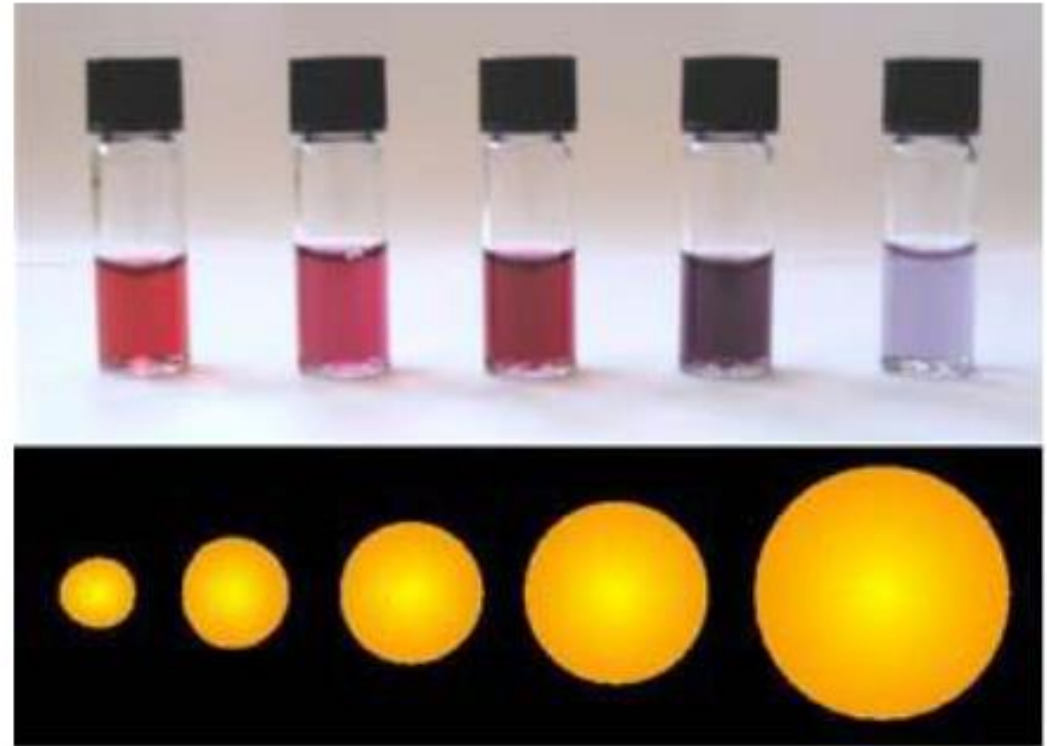
Based on the **surface area to volume effect**, nanoscale materials have:

- a) Increased total surface area.
- b) Increased number of atoms accessible on the surface.
- c) Increased catalytic activity of those large number surface atoms.
- d) Different/tunable surface catalytic properties by the change in shape, size and composition.

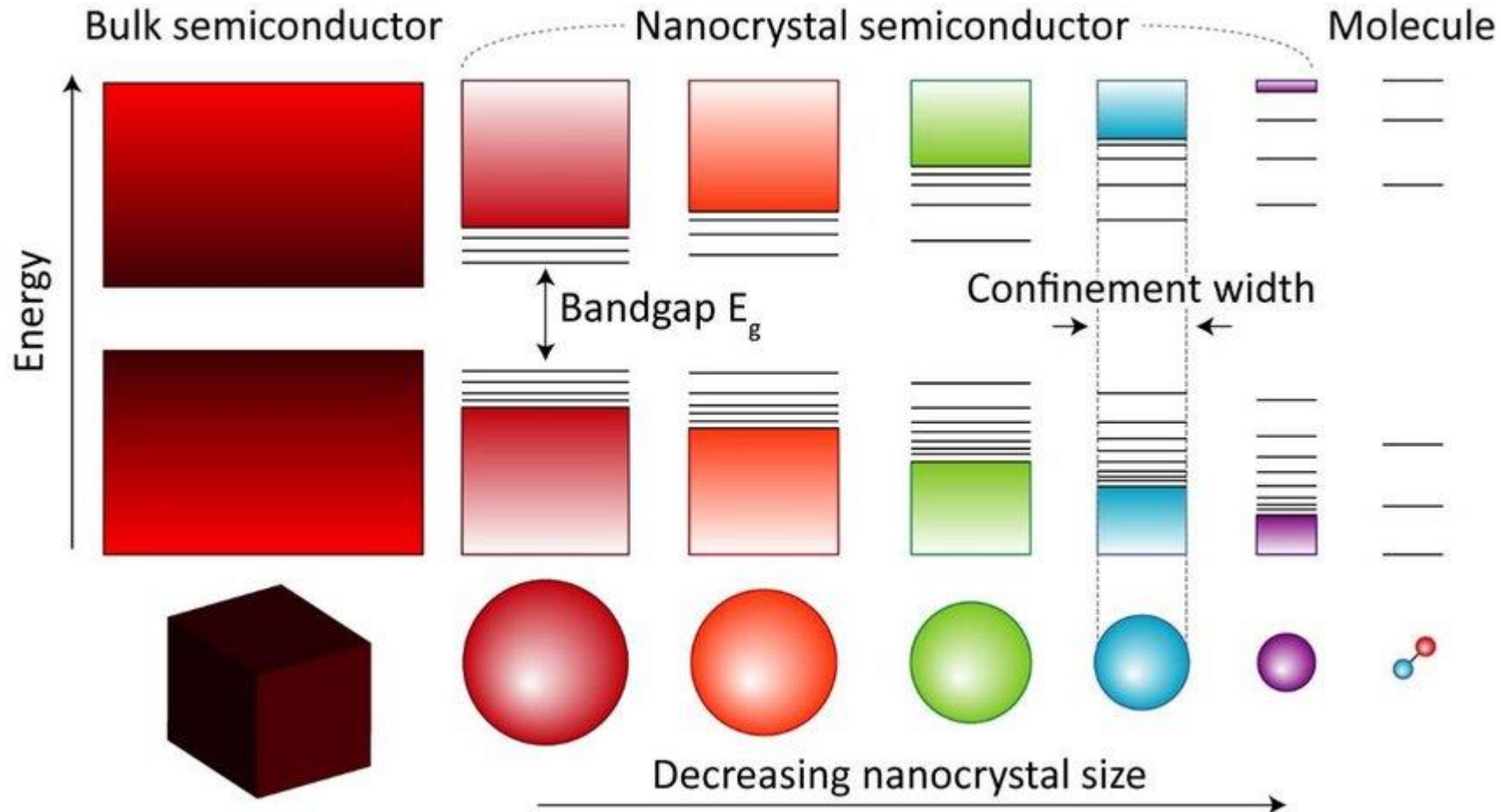
Hence, nanoscale catalysts can increase the rate, selectivity and efficiency of various chemical reactions.

Optical Properties of Nanomaterials

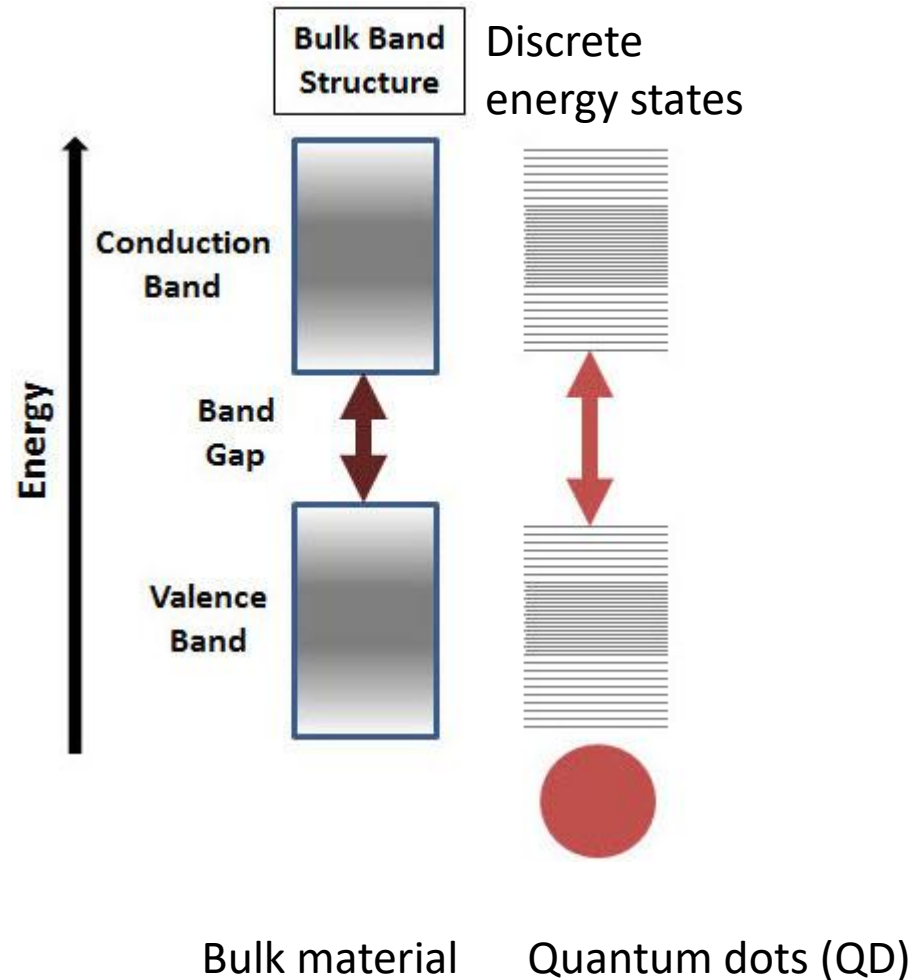
Electromagnetic radiation with matter



Optical Properties of Nanomaterials



Quantum confinements affects the energy levels and electrical conductivity



In bulk materials the electrons are not confined. They are free to move in all Directions to longer distance.

In quantum dots the electrons are confined in all dimension.

Due to electron confinement quantum dots have poor electrical conductivity as compared to bulk material

Quantum dots (QD).

Quantum dots are semicrystalline nanoparticles.

Example: Cadmium Sulphide (CdS) and Cadmium Selenide (CdSe)

They show light absorbance and emission in the visible region of the electromagnetic spectrum.

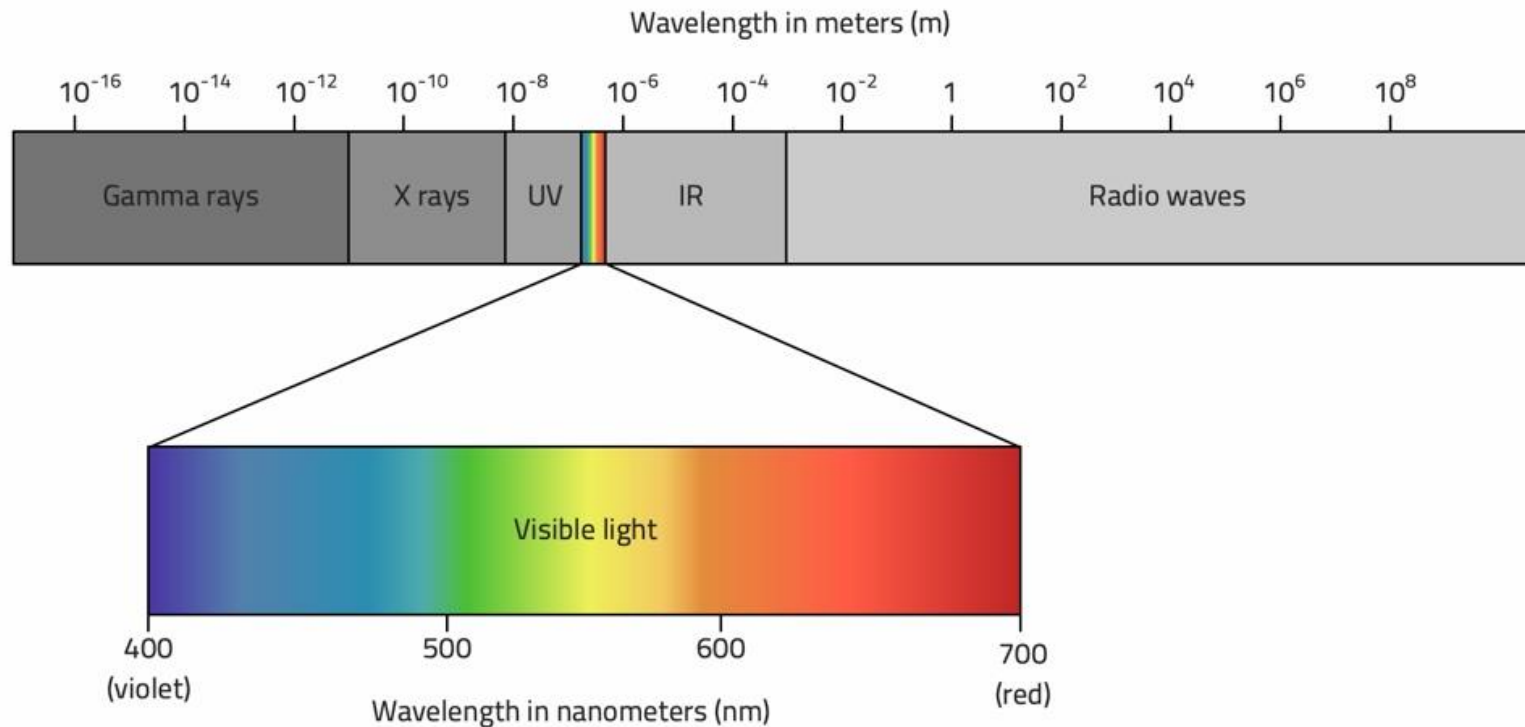
The colour they absorb/emit depends on the size of the nanoparticle.

Optical Properties of Nanomaterials

- Very small sizes metal nanoparticles can develop a **bandgap** (can become a semiconductor or insulator)
- On decreasing the size the electron gets confined to the particle (**confinement effects**) leading to:
 - (i) increase in bandgap energy
 - (ii) band levels get quantized (discrete).

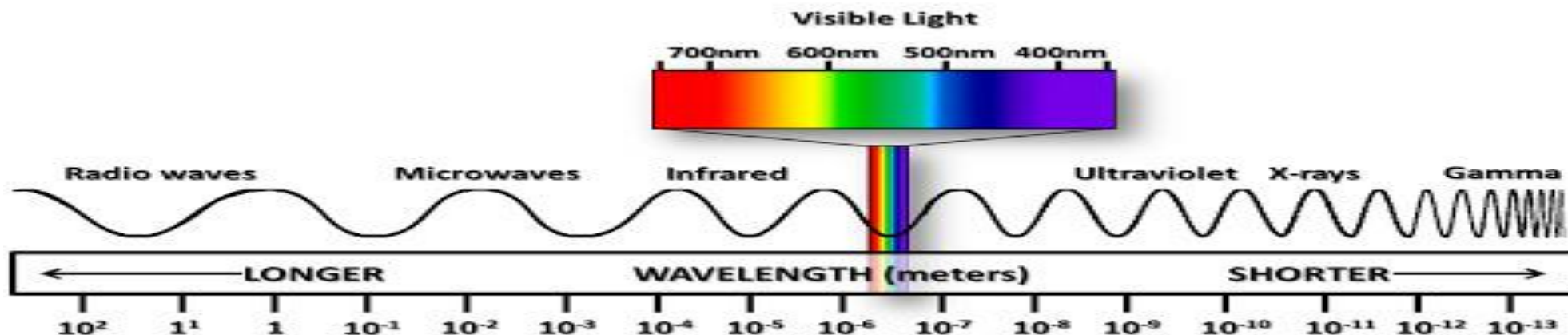
Optical Properties of Nanomaterials

Size increases	Band gap decreases	Emission wavelength increases
Size decreases	Bandgap increases	Emission wavelength decreases



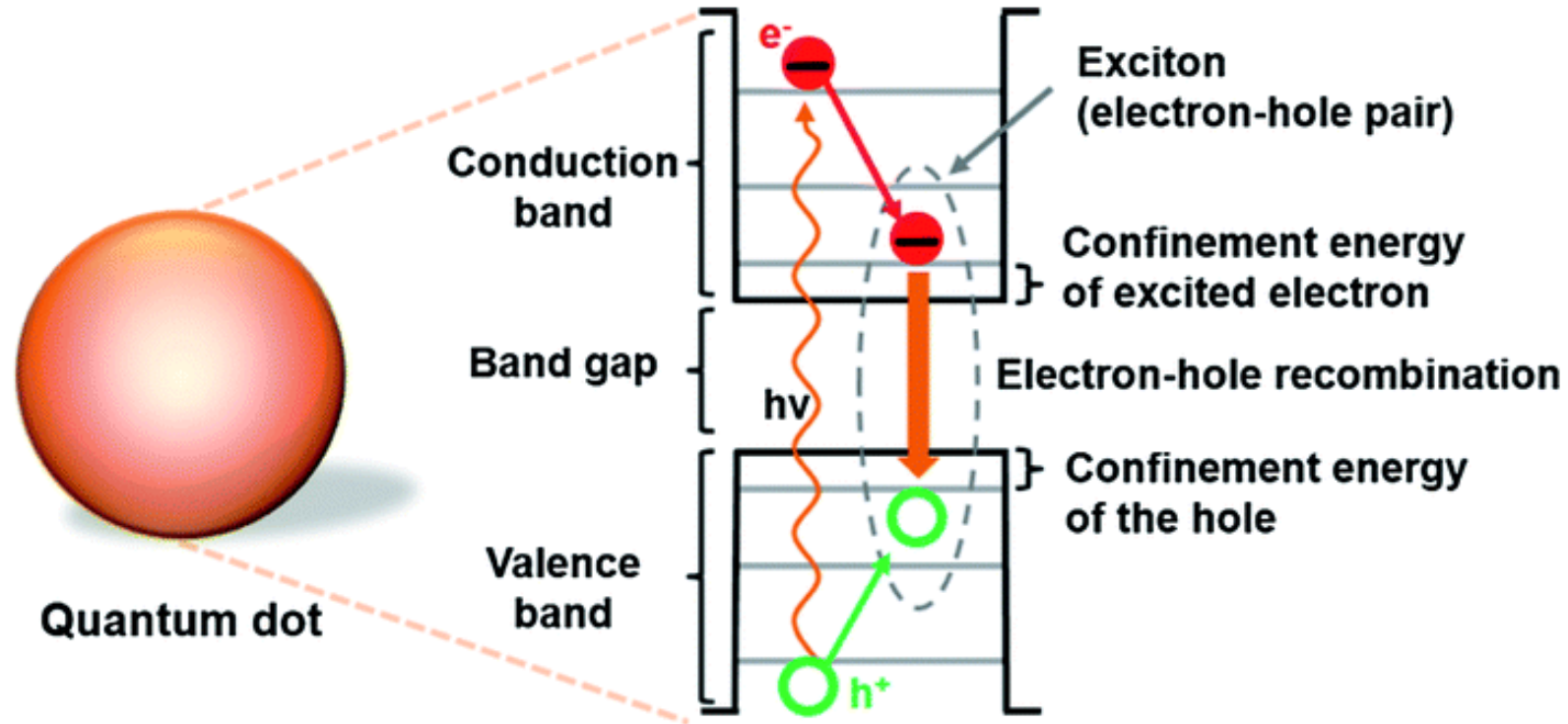
Match the following nanocrystals of different sizes with their emission color

Size of Nanocrystals	Emission Color
1.5 nm	Green
3.25 nm	Violet
5.3 nm	Red
8.1 nm	Orange



Photoluminescence (PL) of quantum dots

Exciton, the combination of an [electron](#) and a positive [hole](#) (an empty electron state in a valence band), which is free to move through a nonmetallic crystal as a unit.



Exciton is formed by the absorbance of light. The electron-hole recombination is accompanied by the emission of light which is called ***fluorescence***.

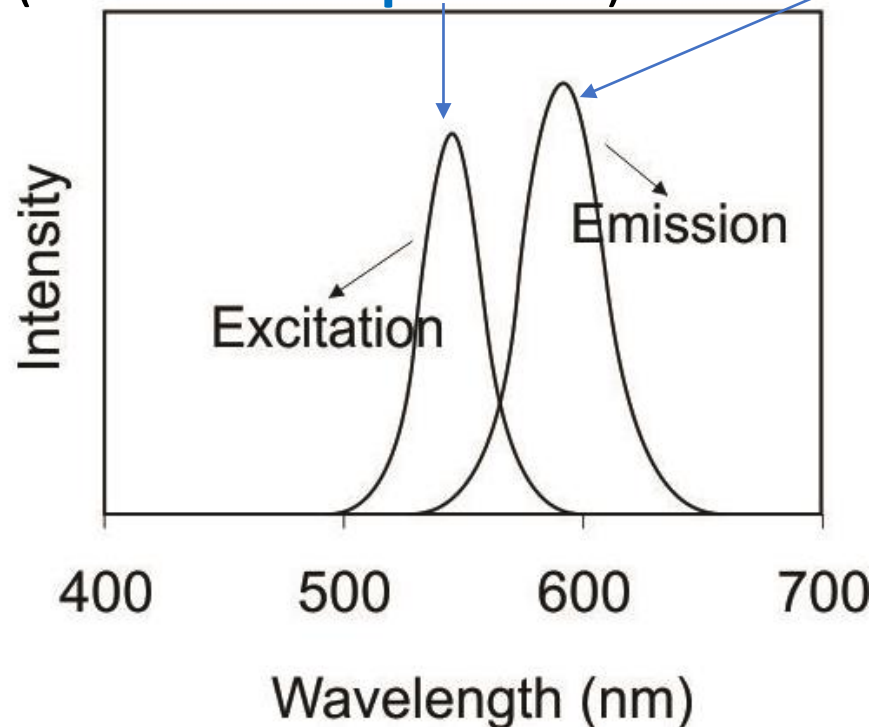
Absorbance and Emission spectrum

The absorption and emission light by material can be quantitatively determined and one can construct a spectrum (***intensity of light as a function of wavelength***).

Input light

(**absorbance spectrum**)

Output light (**emission spectrum**)



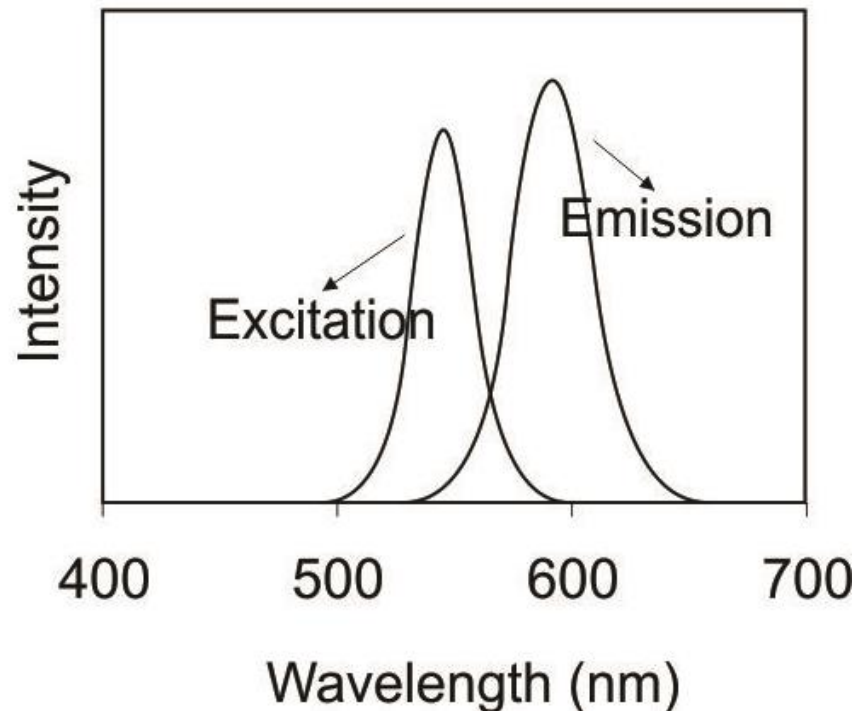
The process of light input to the material is called excitation.

The process of light output from material is called emission.

Absorbance and Emission spectrum

The maxima of emission spectrum (middle wavelength) is always larger than that of absorption maximum.

The emitted light will have less energy than the energy of the light being absorbed. *It is due to the loss of small energy as heat.*



$$E = hc/\lambda$$

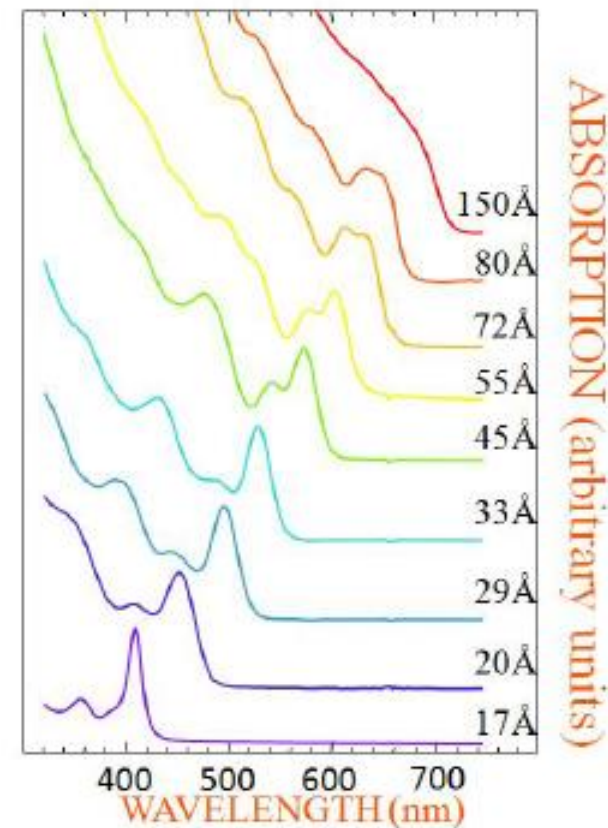
Longer wavelengths
– smaller energy

Smaller wavelengths
– larger energy

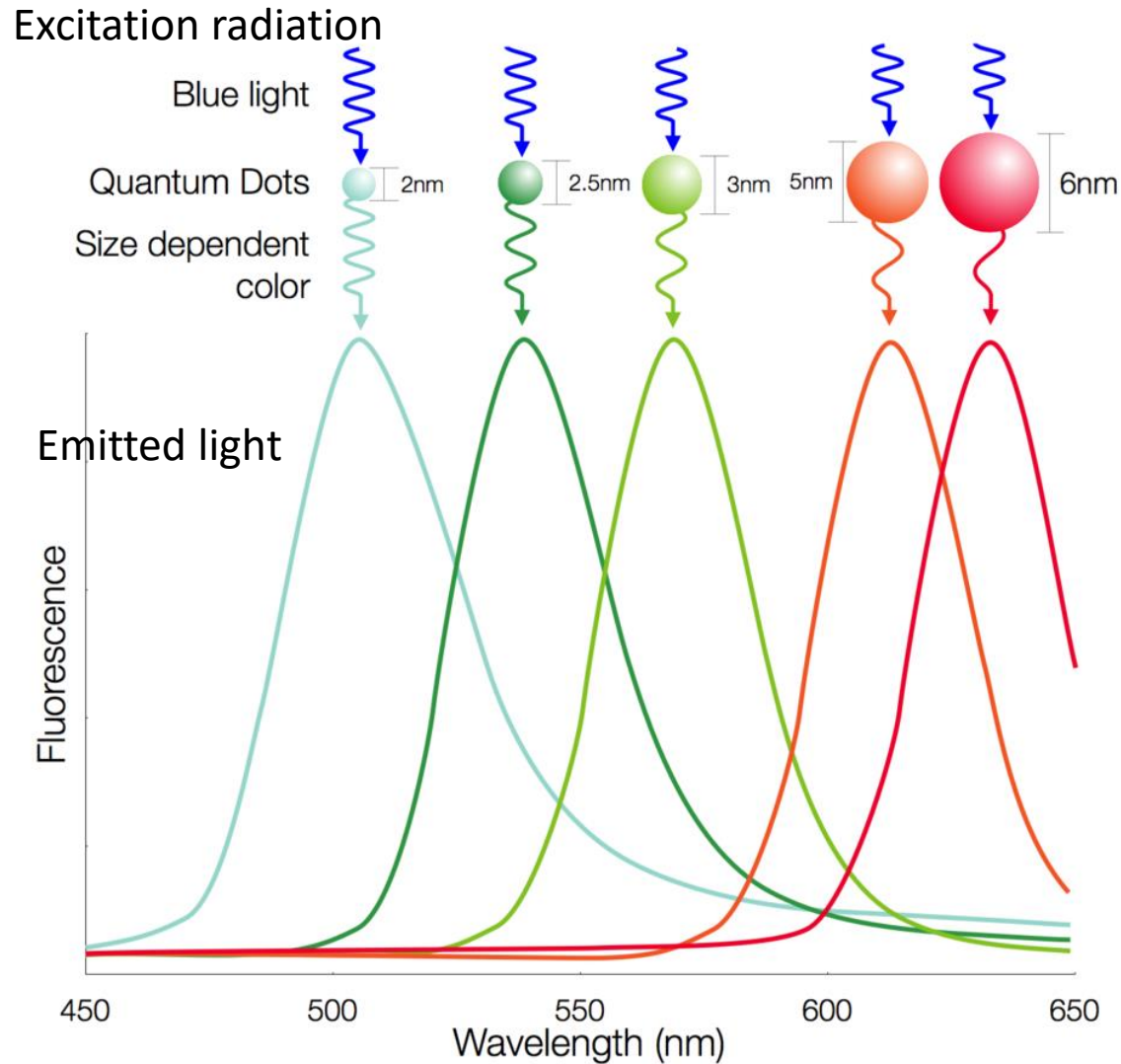
Color of emission and size of the quantum dots

Quantum dots absorb and emit light.

Smaller particle absorbs and emits at *lower wavelength* of the visible radiation.



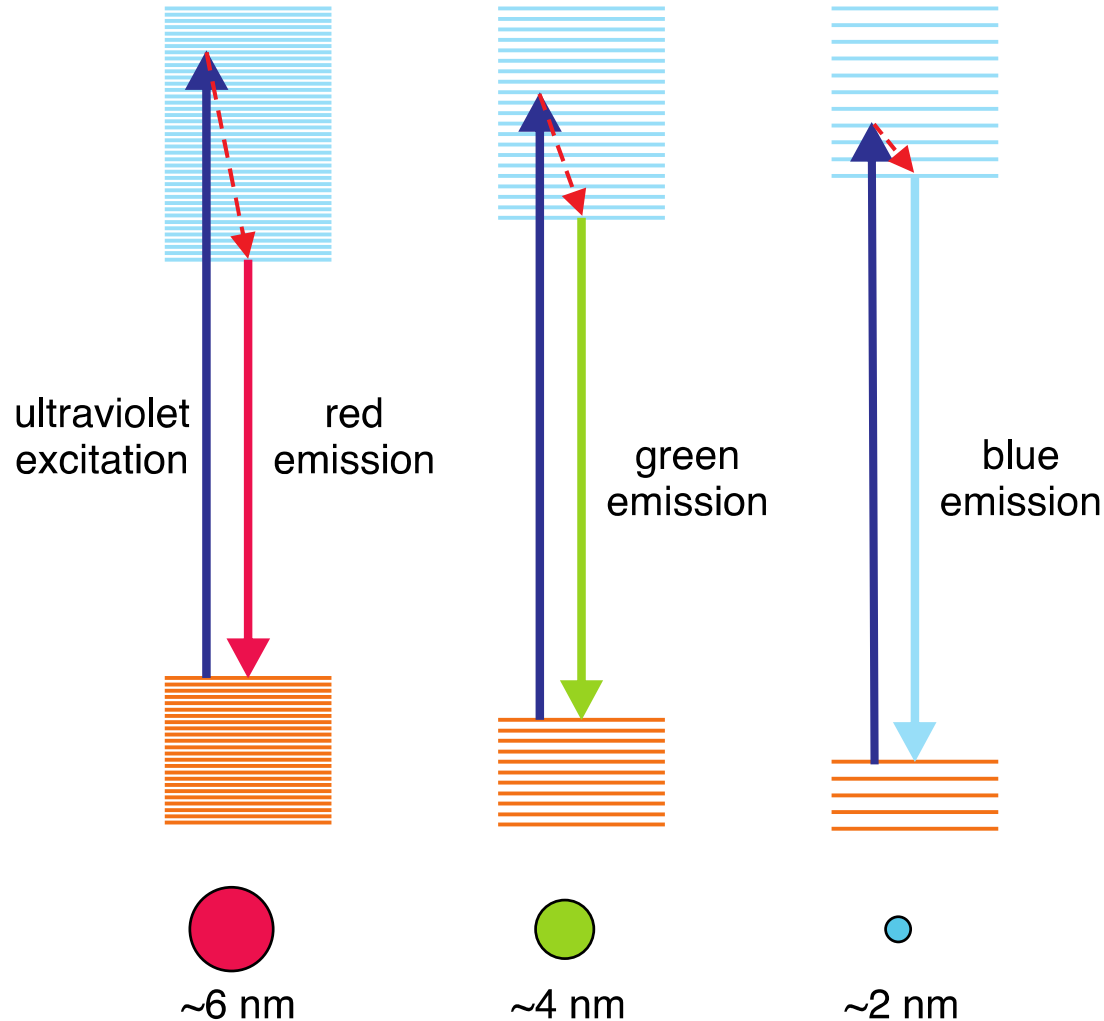
Size dependent emission spectrum of QD



Energy of emitted light (fluorescence) < absorbed light,

So, one can use high energy blue light to excite any sized QD.

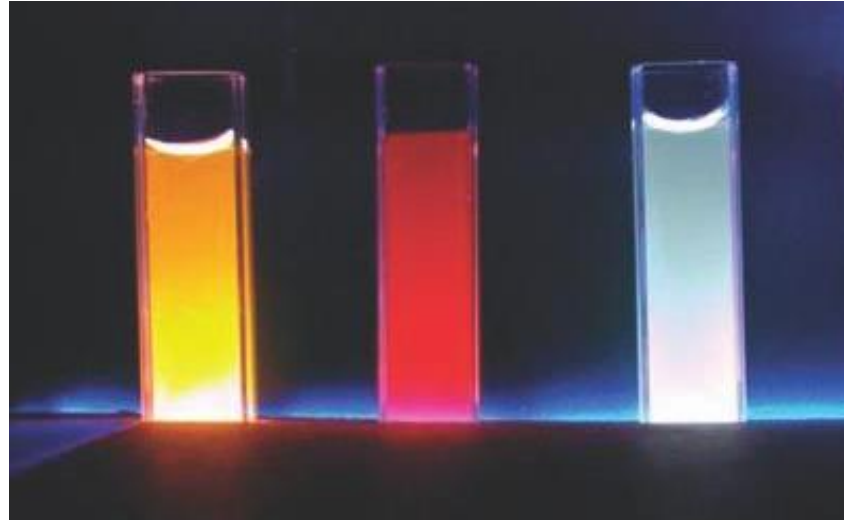
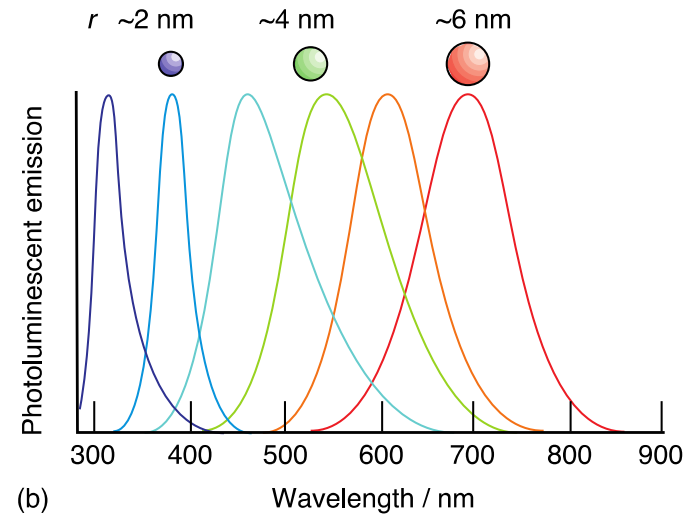
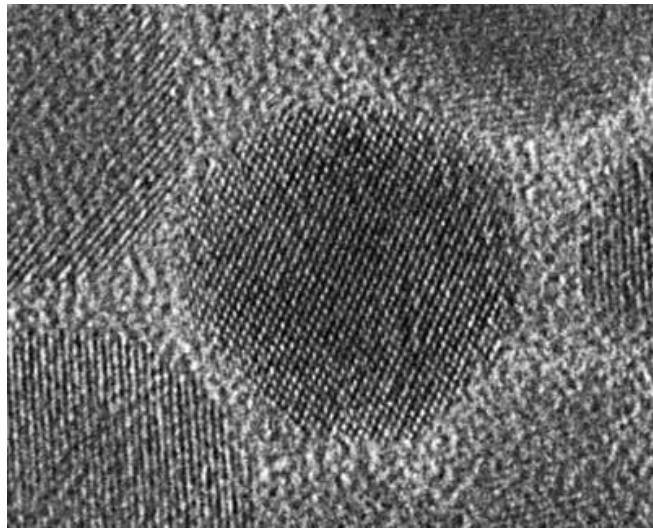
Same absorbance light-Varied emission light



Smaller particle,
larger the band gap.

small wavelength
light is emitted

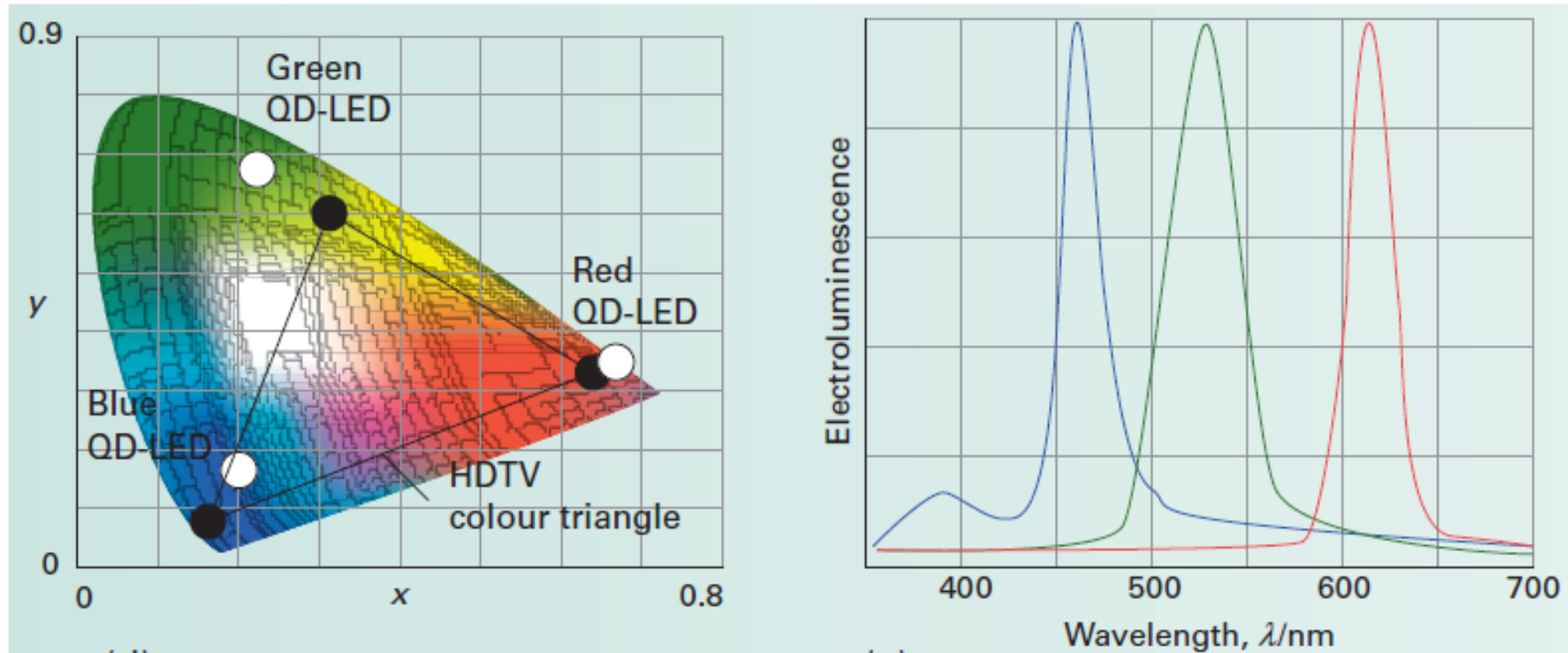
Larger energy is
absorbed or emitted



Cadmium sulfide quantum dots: (a) a CdS dot approximately 8nm diameter; (b) the photo- luminescent colours emitted by CdS quantum dots, schematic; (c) Photoluminescence of two different sized quantum dots (left and centre) and a mixture of the two, giving an approximately white output

Chromaticity diagram shows the positions of red, green, and blue QD-LED colours, an HDTV colour triangle is shown for comparison.

Normalized EL spectra of fabricated QD-LEDs corresponding to the colour coordinates



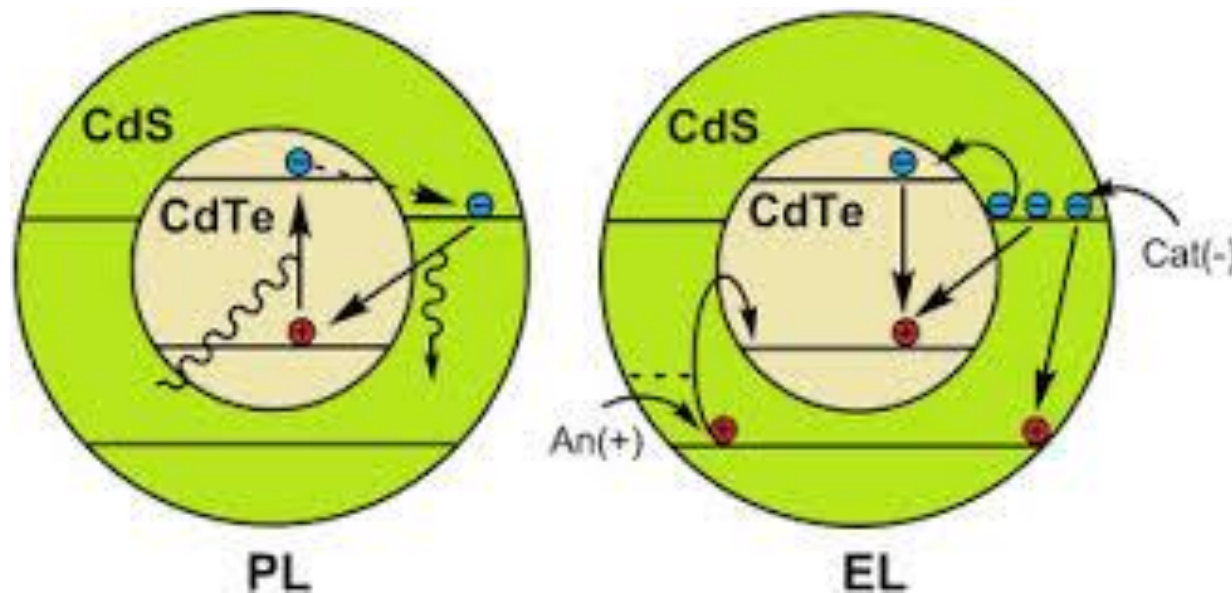


A typical flat-screen TV (left) can only express about 20% of the colors that are detectable with the human eye. A QD-enhanced TV (right) can produce over 60% of the spectrum seen by the human eye.

Reason:

The half-width of the emission spectrum produced by normal LED is wider than the half-width of the emission spectrum produced by the quantum dots (compare the inset figure showing visible region spectrum produced by LED and QD)

Photoluminescence (PL) Vs electroluminescence (EL) of quantum dots



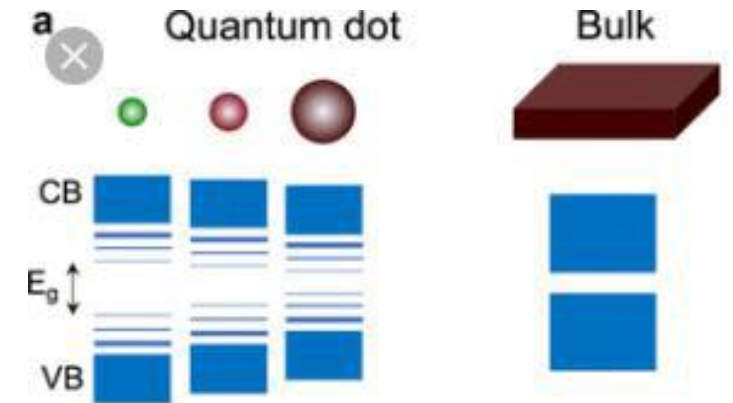
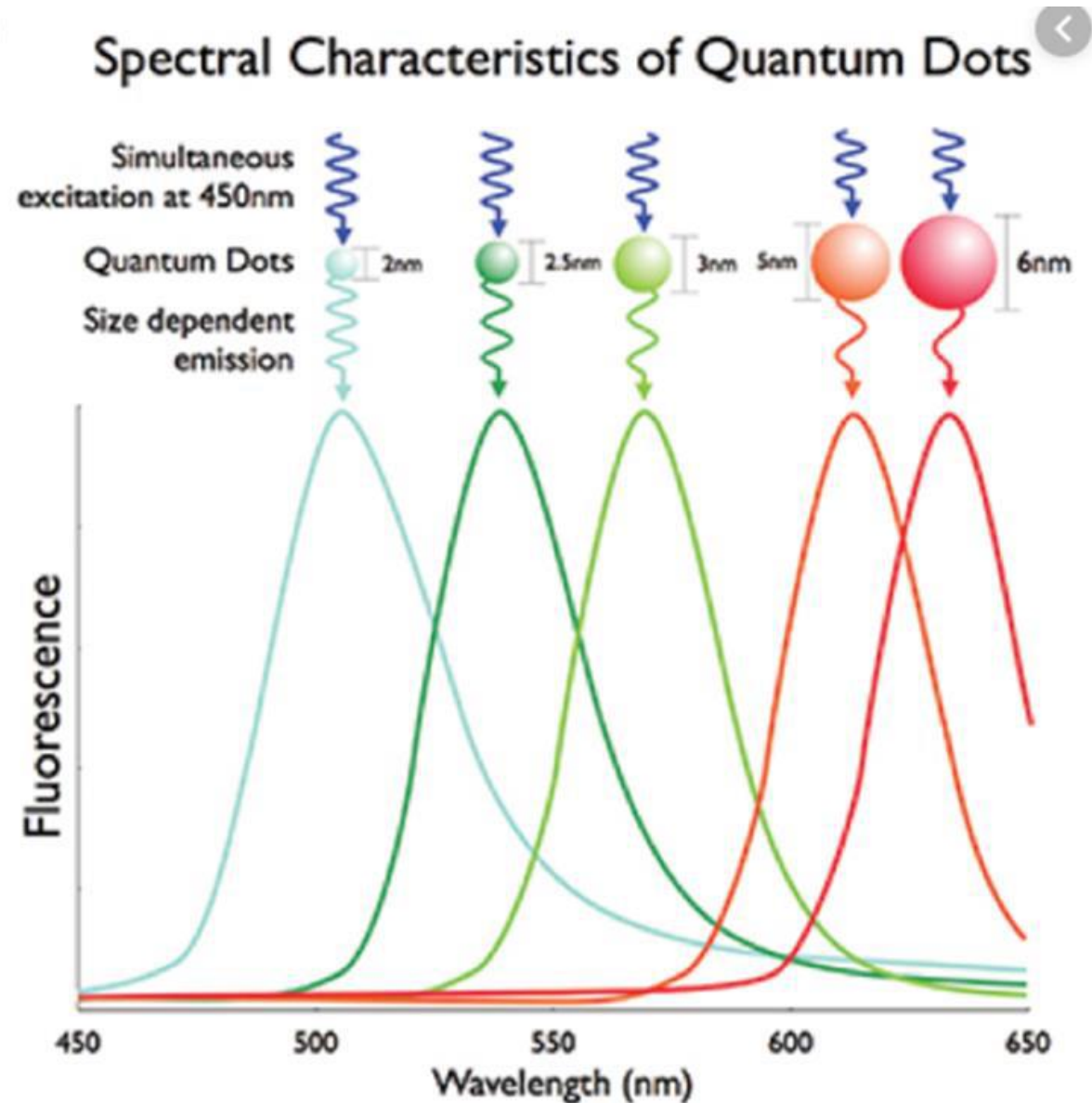
In photoluminescence the **excitons (electron-hole pair)** are produced by the **absorbance of light**.

In electroluminescence excitons are produced by the **application of potential bias**.

Electronic Properties of Nanomaterials

- In **bulk materials**, conduction of electrons is **delocalized**, that is, electrons can move **freely in all directions**.
- When the scale is reduced to **nanoscale**, the **quantum effect** dominates. For **zero dimensional** nanomaterials, all the dimensions are at the nanoscale and hence the electrons are confined in **3-D space**. Therefore **no electron delocalization** (freedom to move) occurs.
- For **one dimensional** nanomaterials, electrons confinement occurs in **2-D space** and hence electron delocalization takes place along the axis of nanotubes/nanorods/nanowires.
- Due to electron confinement, the energy bands are replaced by discrete energy states which make the conducting materials to behave like either semiconductors or insulators

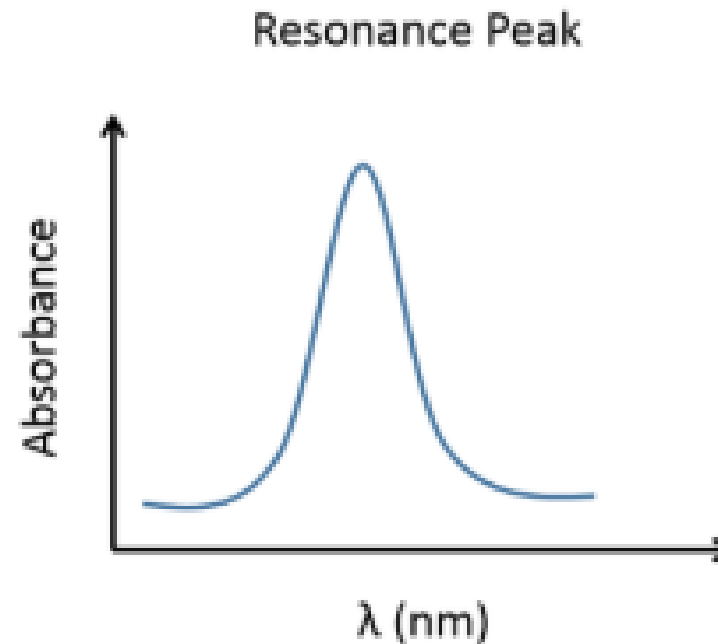
Semiconducting Nanomaterials



Larger particle emitting longer wavelength light

Metallic nanoparticles

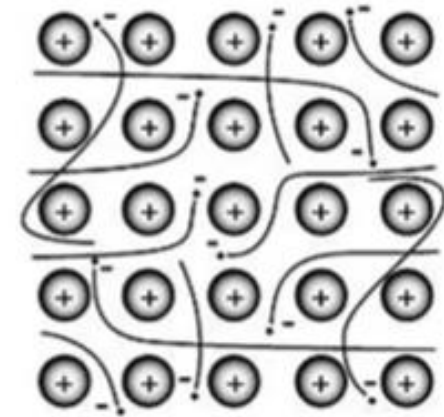
- A prominent spectroscopic feature of noble metal nanoparticles (NPs) is **surface plasmon resonance**, which gives rise to a **sharp and intense absorption band** in the visible range.



Surface plasmon resonance

- All metals have free electron in valance orbit
- The free electrons in the material naturally **resonate** at a **frequency** that depends on the **composition, size and shape** of the particle

When the wavelength of incident light matches the oscillation frequency, the particles can **strongly absorb** or scatter the light resulting in a **strongly colored particle**.

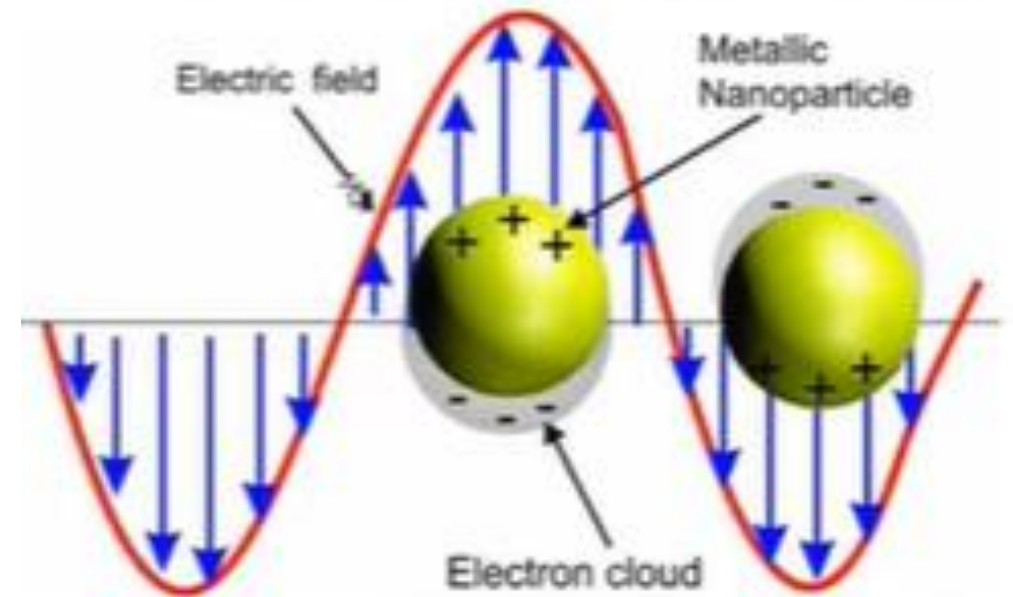


Electron sea

Surface plasmon resonance

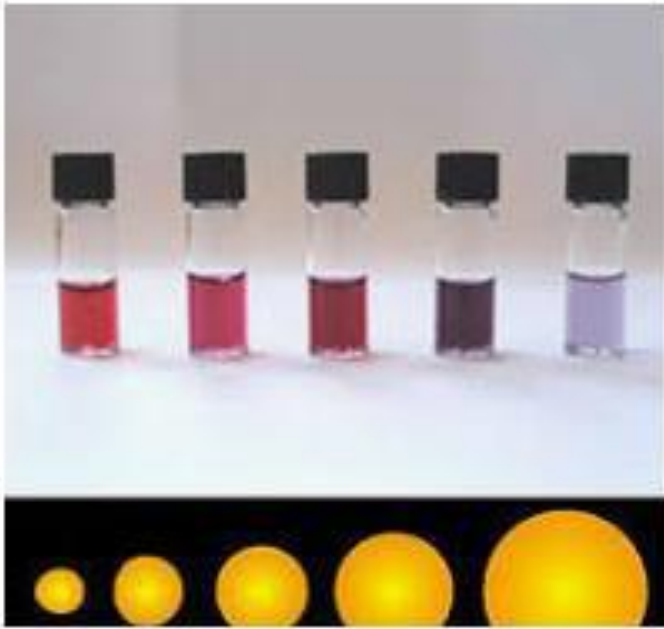
The physical origin of the absorption is a collective resonant oscillation of the free electrons of the conduction band of the metal

At a specific wavelength (frequency) of light, collective oscillation of electrons on the metal nanoparticle surface cause surface plasmon resonance resulting in **strong extinction of light** (absorption and scattering)

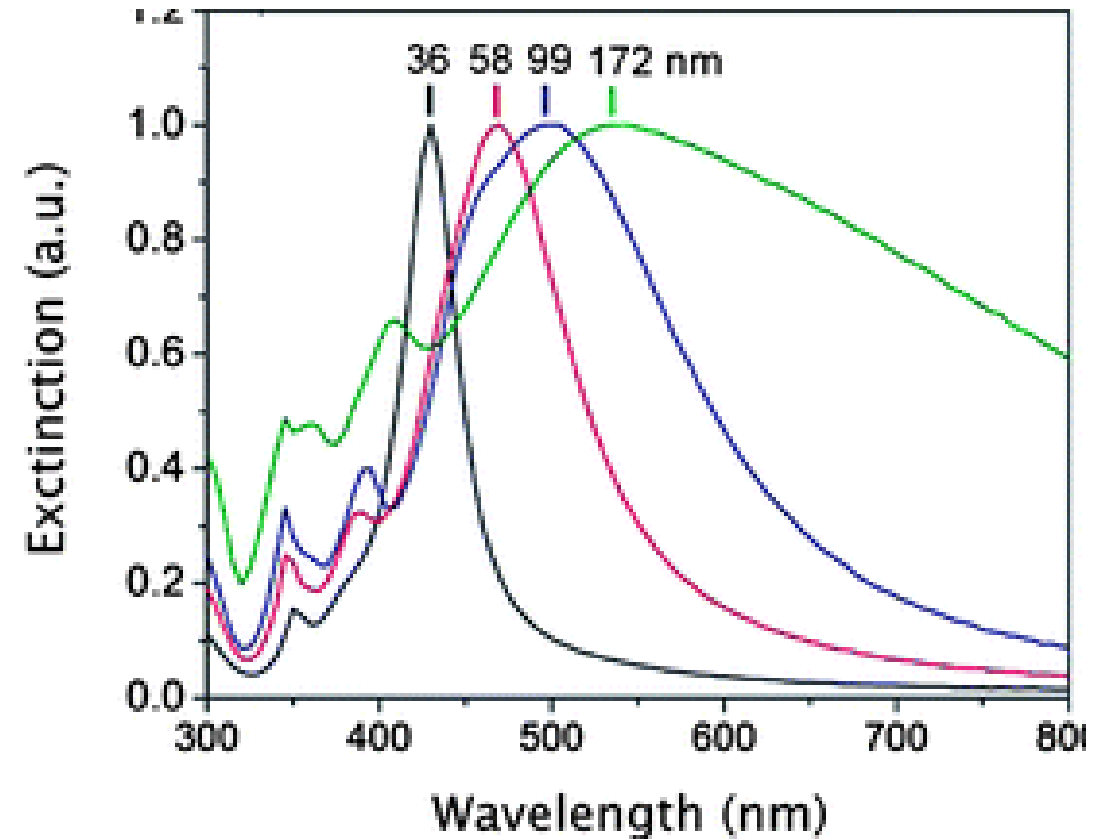


The wavelength of the light are larger than the size of particle !!!

Surface plasmon resonance

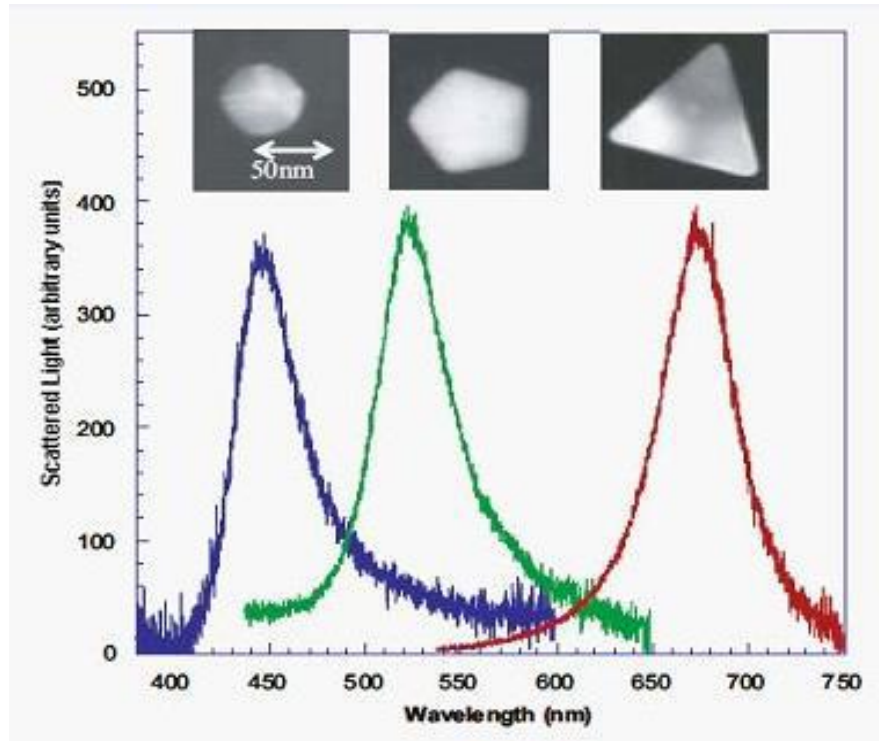


Surface plasmon resonance wavelength depends on **size of the particle**

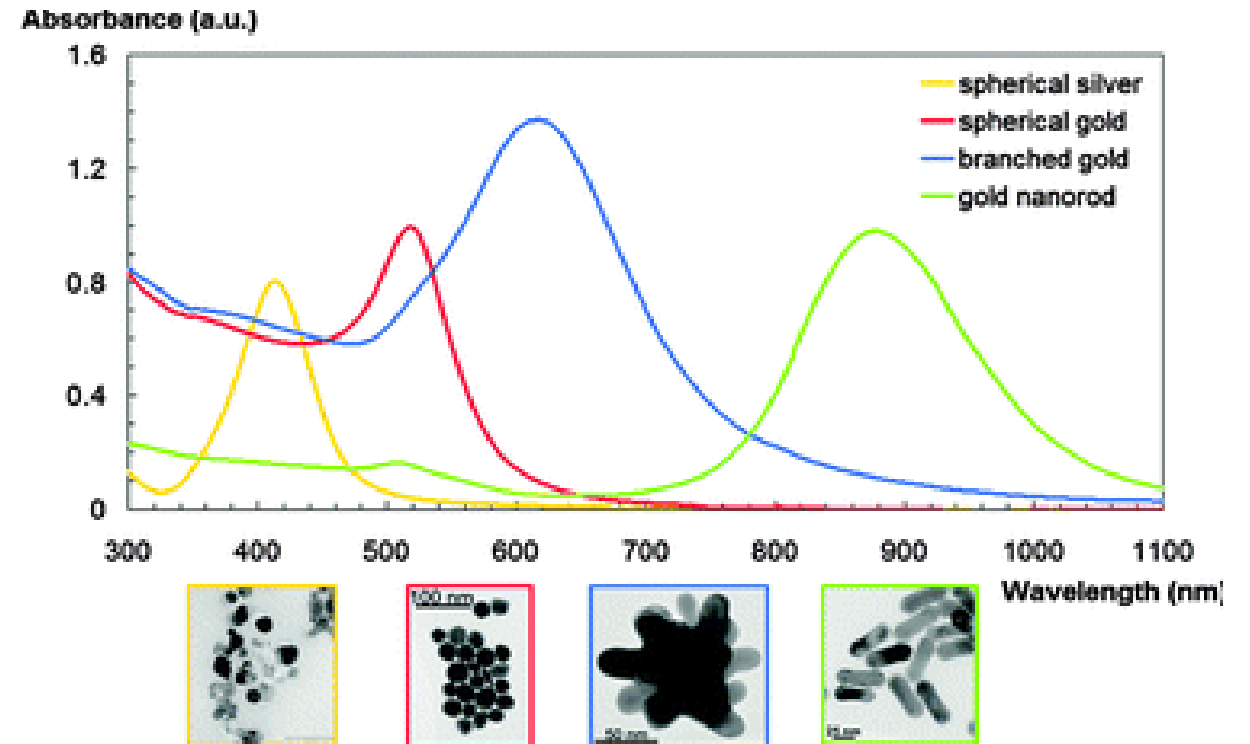


if particle size is larger, surface plasmon resonance appears at longer wave length

Surface plasmon resonance

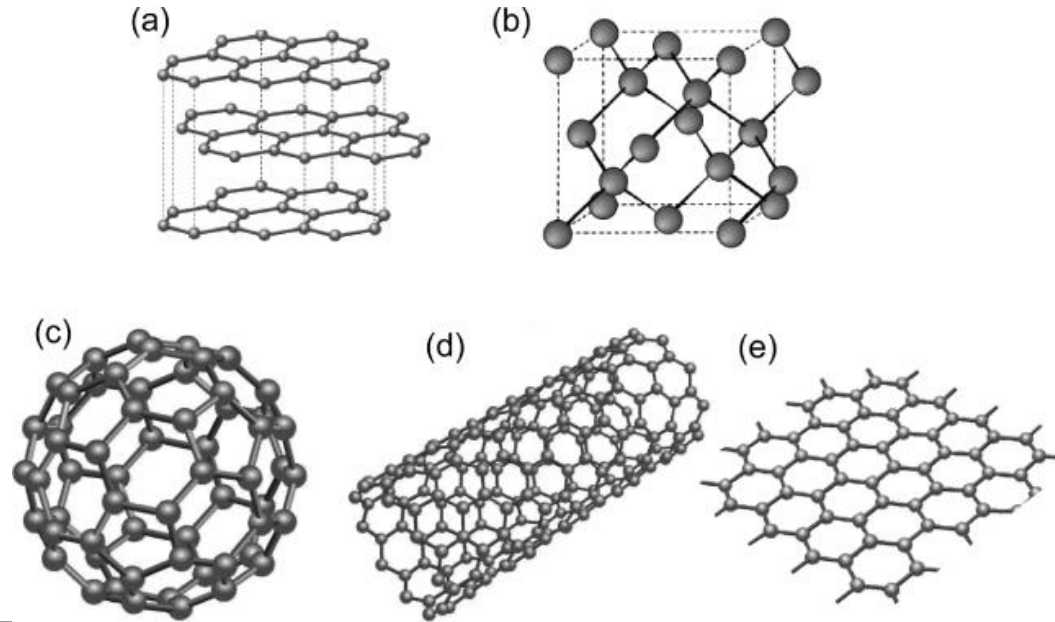


Surface plasmon resonance wavelength depends on **shape of particle**



Different shapes show different surface plasmon resonance peaks

Carbon Materials



Forms of carbon:

- a) Graphite 3D
- b) Diamond 3D
- c) Buckminsterfullerene
"Buckyballs" 0D
- d) Carbon nanotube 1D
- e) Graphene 2D



Andre Geim



2010 Nobel Prize in Physics

http://www.nobelprize.org/nobel_prizes/physics/laureates/2010/



Konstantin Novoselov

Properties of graphene

Thinnest material possible (~0.345 nm thick)

Strongest material ever measured

- 200 times more stronger than steel

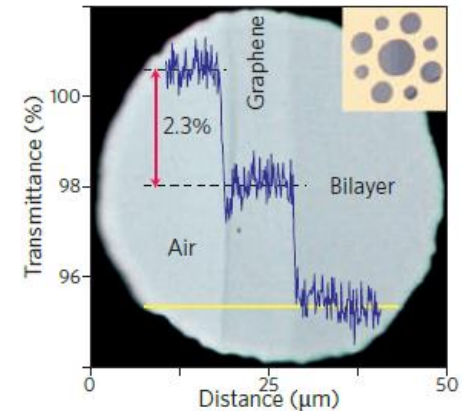
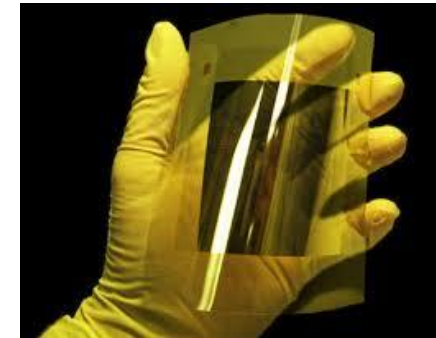
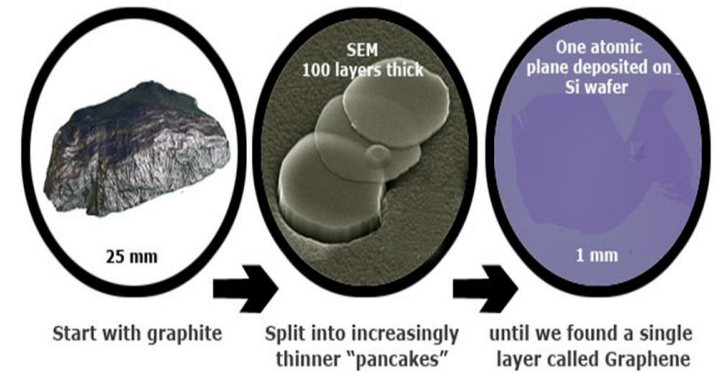
Electrical conductivity is best known

so far for any material

- 10,00,000 times more conductive than copper

Flexible and stretchable

Transparent (~97.7% white light can pass through a single layer of graphene)



Potential applications of graphene

Optoelectronics

LEDs, Touch screens,
Flexible displays etc.

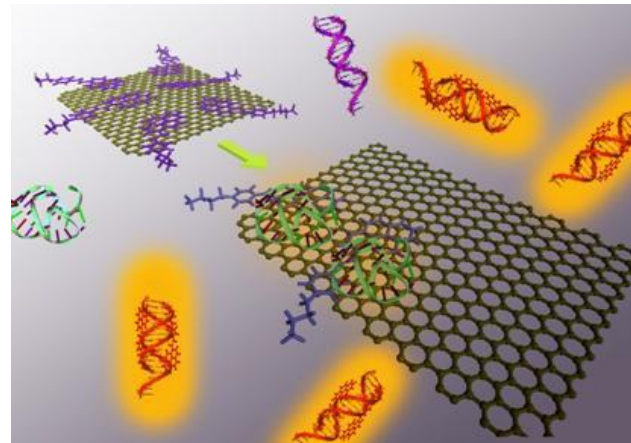


Energy generation and storage

Photovoltaics,
Supercapacitor etc.

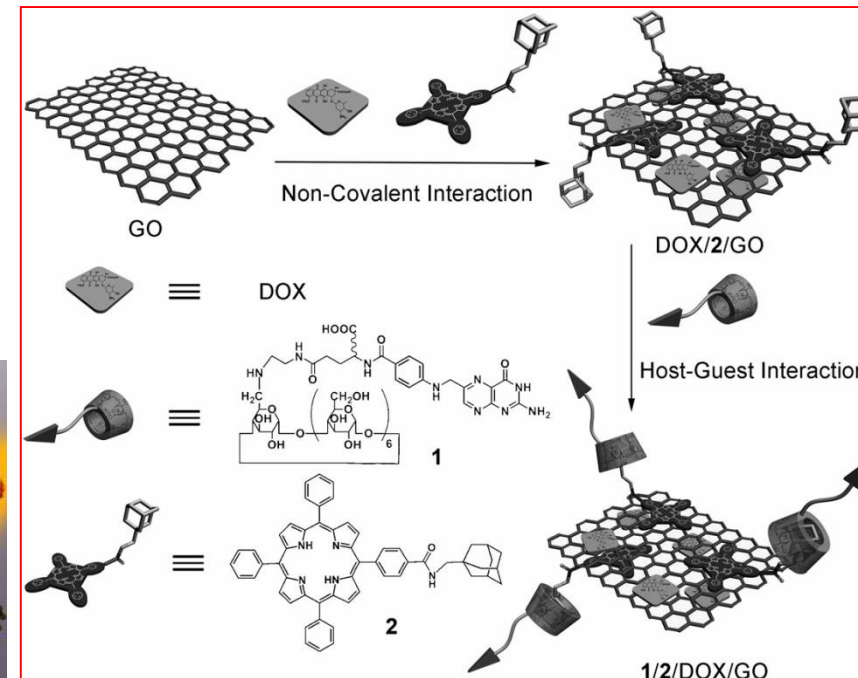
Drug delivery

Anticancer
drug: Doxorubicin



Sensors

DNA sensor



Chem. Eur. J. 2012, 18, 4208

Angew. Chem. Int. Ed. 2010, 49, 6549

Questions

Energy conversion

1. The wave length of the emitted radiation is 468 nm, calculate the band gap energy of the material in kJ/mol

Solution 1

$$E_{\text{photon}} = \frac{h \cdot c}{\lambda}$$

The number of particles in one mole is given by the Avogadro constant, which is $N_A = 6.02 \cdot 10^{23} \text{ mol}^{-1}$. In this case the particles are photons.

As we have the expression for energy per photon, we need to multiply it by the Avogadro constant to get the amount of energy per one mole of photons.

$$E_{\text{per mole}} = \frac{N_A \cdot h \cdot c}{\lambda}$$

Note how the units for this expression cancel out to what we desire:

$$\frac{J \cdot s \cdot m}{s \cdot m \cdot \text{mol}} = \frac{J}{\text{mol}}$$

Using SI units (m , $\frac{m}{s}$, $J \cdot s$, mol^{-1}), substitute all the values into the equation to get the answer:

$$\begin{aligned} E_{\text{per mole}} &= \frac{6.02 \cdot 10^{23} \cdot 6.63 \cdot 10^{-34} \cdot 3.00 \cdot 10^8}{468 \cdot 10^{-9}} \\ &= 256 \cdot 10^3 \frac{\text{J}}{\text{mol}} \end{aligned}$$

Finally, convert joules to kilojoules:

$$E = 256 \frac{\text{kJ}}{\text{mol}}$$

Example 2. Convert 2eV to nm

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

$$E = \frac{hc}{\lambda}$$

$$\lambda = \frac{hc}{E}$$

The wavelength of a 2 eV photon is given by:

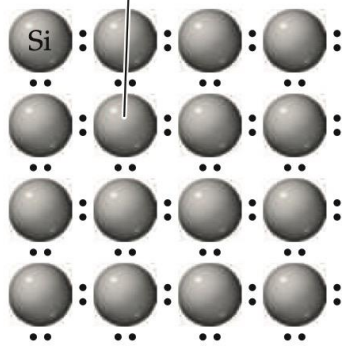
$$\lambda = hc / E_{ph} = 6.625 \times 10^{-34} \times 3 \times 10^8 / (1.6 \times 10^{-19} \times 2) = 621 \text{ nm.}$$

3. Identify each of the following as a p-type or n-type semiconductor:

- (a) aluminum doped silicon,
- (b) antimony doped germanium,
- (c) boron doped silicon,
- (d) Gallium doped germanium
- (e) Boron doped diamond

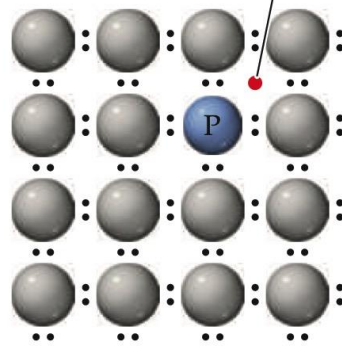
4. Illustrate the band structures of i, n, and p-type semiconductors

Silicon forms bonds to four other silicon atoms and has a complete octet.



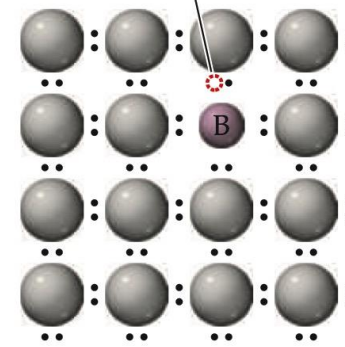
Si crystal

A silicon crystal doped with a P atom has an extra electron that is not needed for bonding. The *negative* electron is the charge carrier.



n-type semiconductor

A silicon crystal doped with a B atom is electron deficient. The charge carrier is called a *positive* hole (represented by an empty circle.)



p-type semiconductor



An *n*-type semiconductor, such as silicon doped with phosphorus, has more electrons than needed for bonding and thus has more electrons in the partially filled conduction band than pure silicon.



Conduction band
(antibonding)

Valence band
(bonding)

A *p*-type semiconductor, such as silicon doped with boron, has fewer electrons than needed for bonding and thus has more vacancies—positive holes—in the valence band than pure silicon.



5) If Ge crystals has been doped with arsenic, it will behave as n-or p-type semiconductor?

Demonstrate the change with band theory with the help of drawing band structures.

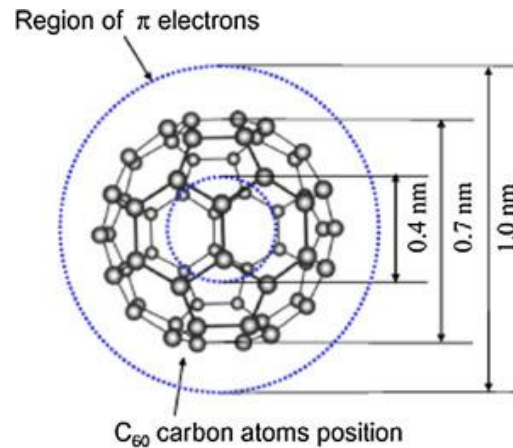
6) What is Exciton and how it is formed?

7) Why conductivity of semiconductor increases with increase in temperature?

8) Why conductivity of metal decreases with increase in temperature?

9) What determines the colour of the emitted light by the quantum dots?

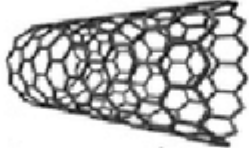
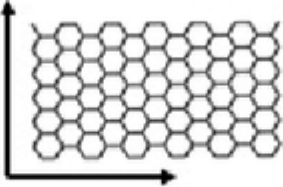
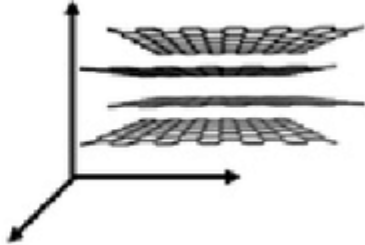
What is the macroscopic dimension of Buckminster fullerene (C_{60})?



Fullerene is a carbon allotrope. It is made up of 60 carbon atoms. All carbon atoms and its π -electron cloud is confined within 1 nm. It does not expand to micro or even bigger macro length scale. So, ***it is a 0D material at macroscopic scale*** (its electron cloud is absent beyond 100 nm in all dimension)

However, its dimension (**or electron confinement**) is 3D at nanometer scale (because its electron cloud is present in all directions within 100 nm from the center of the ball)

Classify fullerene, carbon nanotubes, graphene and graphite according to their macroscopic dimension?

0D	1D	2D	3D
			
Fullerene	Carbon Nanotube	Graphene	Graphite

All dimension the
electron is confined

In no direction the
electron is confined